ProveIt research continuation

# Lattice Contacts and Mixed Zeta Identities

Unequal harmonic tails, shifted Stieltjes products, and Gamma spectral faces

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Prepared for the ProveIt project

**Abstract.** We continue several explicit research questions in the ProveIt manuscript and its latest incoming reports. A single elementary generating function gives every contact term at minimally compensated nonpositive even orders of a vertical Hurwitz lattice; periodic weights remove all these terms exactly when their mean is zero. An order-dependent Mellin identity supplies exact modular transport and every nonmodular derivative correction. A polarized Bernoulli identity extends the centered-moment calculus to arbitrarily many unequal even-order Hurwitz tails, with terminating recurrences, finite multiple-zeta spans, and explicit six- and eight-coordinate examples. Independently shifted products of generalized Stieltjes constants are evaluated by a jointly holomorphic generator with geometrically convergent Hurwitz series. Finally, every polynomial spectral-face coefficient is explicit, every Gamma face with one displaced mass reduces to depth-two polylogarithms, and the precise polar obstruction for independently variable spectral orders is classified and removed. All claimed identities have analytic or finite algebraic proofs. Numerical checks serve as independent diagnostics. A source-status correction, integration guidance, and a prioritized research agenda are included.

Source snapshot: `36edefa8c531d9d6df8306ac51bcc317250f404c`

Novelty is assessed against the inspected corpus and specified primary literature; historical priority is not asserted.

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# 1 Research frontier, principal identities, and conventions

This article develops exact identities at the meeting point of Hurwitz continuation, polynomially weighted harmonic tails, Gamma products, and polylogarithms. The source is the ProveIt repository at commit `36edefa8c531d9d6df8306ac51bcc317250f404c`. The manuscript and the ten incoming archives present at that snapshot were used to identify the current frontier [1, 2]. The proofs below are self-contained apart from standard special-function identities and explicitly credited predecessor formulas that are restated before use.

Two short consequences convey the character of the results. Put

$$x_n = n + \tfrac{1}{2}, \qquad T_p(n) = \zeta(p, n+1) = \zeta(p) - H_n^{(p)}.$$

Then

$$\boxed{\sum_{n=0}^{\infty} T_2(n)T_4(n)(1 - x_n T_2(n)) = \frac{5}{16}\zeta(6),} \tag{1.1}$$

and

$$\boxed{\begin{aligned}\sum_{n=0}^{\infty} T_2(n)^2 T_4(n) \left\{ \left(x_n^2 + \frac{1}{12}\right) T_2(n) - x_n \right\} \\ = \frac{59}{12}\zeta(3)\zeta(4) + \frac{23}{6}\zeta(2)\zeta(5) - \frac{203}{16}\zeta(7).\end{aligned}} \tag{1.2}$$

Both sums are ordinary and absolutely convergent. Their summands decay respectively as $O(x_n^{-6})$ and $O(x_n^{-8})$. They follow from the same finite recurrence that computes all regular centered moments of any product of positive even-order tails; see Corollaries 5.2 and 5.4. No finite-part convention is involved in these two formulas.

## 1.1 What is proved

1. **Every minimal lattice contact.** At $s = -N$, $N$ even, the $d$-th order derivative of the minimally compensated Hurwitz lattice has a complete polynomial in $\log(hK)$, where $K$ is the integer cutoff. Its generating function is

$$\frac{\tau\beta_N}{2}\, u \prod_{j=1}^{N}(1 - u/j)\frac{\sin(\pi u/2)}{\pi u/2} e^{-Xu}, \qquad \tau = \frac{2\pi}{h}, \quad \beta_N = \frac{B_{N+2}}{(N+1)(N+2)}.$$

   The coefficient multiplied by $d!$ is the contact of order $d$. This answers the all-derivative and filter questions in the vertical-lattice report [3].

2. **Exact order-dependent modular transport.** The Lambert family $\Lambda(s, \tau)$ in (3.1) has a holomorphic contour transformation near every nonpositive even order and near $s = 1$. An explicit trigonometric polynomial recurrence produces all derivative corrections. The zeroth-order eta and Ramanujan transformations are credited classical specializations; the higher derivatives retain specified convergent contour terms.

3. **Arbitrarily many unequal even tails.** For a finite labeled collection of even integers $p_i \geq 2$, every regular centered moment of $\prod_i T_{p_i}$ is computed by a finite recurrence on proper subsets. Its

generating function is analytic for $|t| < 2\pi$. All moments lie in a fixed rational span of convergent multiple zeta values of weight at most

$$\sum_i p_i - 1 - \mathbf{1}_{\{|I| \text{ odd}\}}.$$

The mixed products $T_2^2 T_4$ and $T_2^3 T_4$ have explicit generators and full rank on six and eight specified formal coordinates. Their first six odd and first eight even moments, respectively, reconstruct every later regular moment.

4. **Independent nonperiodic Stieltjes shifts.** For $\operatorname{Re} a, \operatorname{Re} b > 0$, the integral

$$\int_0^1 \left(\zeta(1+u, a+x) - \frac{1}{u}\right)\left(\zeta(1+v, b+x) - \frac{1}{v}\right) \mathrm{d}x$$

is evaluated by a finite incomplete-beta prefix, explicitly removable Gamma quotients, and two geometric Hurwitz series. The expression is jointly holomorphic in both orders. Every product of generalized Stieltjes constants and their argument derivatives follows by coefficient extraction. Ordinary polygamma products admit a stronger terminating reduction to two logarithmic Hurwitz sums.

5. **All spectral-face coefficients and a depth-two collapse.** A finite formula in harmonic star sums gives $I_k(B \mapsto B^m; b_1, \dots, b_k)$ for every $k, m$. With the exact face normalization of Section 7, one has

$$\begin{aligned} &I_k(G_a; \underbrace{B, \dots, B}_{k-1}, B+\delta) \\ &\quad = (-1)^{k-1}(k-1)! \sum_{n=0}^{\infty} \operatorname{Li}_{1,k-1}\left(\frac{\delta}{(n+a)^2 + B + \delta}\right), \end{aligned}$$

where $k \geq 2$, $a > 0$, $B, \delta \geq 0$, and $G_a(B) = \log\Gamma(a + \mathrm{i}\sqrt{B}) + \log\Gamma(a - \mathrm{i}\sqrt{B}) - \log(2\pi)$. Thus one displaced mass has functional depth two in every simplex dimension, with no small-displacement restriction.

6. **The joint-order obstruction and its exact correction.** For polynomially weighted products of quadratic spectra, the only possible polar divisor through the origin is $\sum_i \alpha_i = 0$. Its numerator is an explicit finite polynomial. For unequal masses and no periodic multiplier, the joint germ is holomorphic exactly when the polynomial weight $W(x)$ is even. A mean-zero periodic multiplier removes the obstruction for every polynomial. After the specified local subtraction, all mixed derivatives have convergent simplex formulas.

## 1.2 Relation to established work

The unweighted product sums of zeta tails already have finite multiple-zeta evaluations; Hoffman gives a general treatment [9]. Later product-sum work includes Li and Chu [10]. Our new tail assertion concerns the complete centered polynomial-moment system for arbitrary unequal even orders, including its subtraction constants and anchored analytic generator. The displayed short identities are useful consequences of that larger system.

The incomplete-beta method for products of Hurwitz zeta functions has a direct predecessor in Shpot and Paris [11], and logarithmic digamma-square integrals were studied by Connon [12].

The additional content here is the simultaneous treatment of two independent ordinary argument shifts, joint removal of both order poles, and an explicit derivative calculus with proved normal convergence. Likewise, Dirichlet averages and Gamma products are classical [13, 8]. They underlie the new collision coefficients and displaced-mass evaluation rather than constitute novelty claims.

Vertical Gamma-product identities also predate this continuation, including recent work of Arabi Ardehali and Rosengren [14]. Generalized Stieltjes modular relations are established in the work of Dixit, Sathyanarayana, and Sharan [15]. The present lattice and modular sections specify the all-order boundary polynomials, their cutoff coordinates, and the exact contour corrections for this particular sampling family. The literature comparison is targeted. It supports a continuation of the inspected ProveIt questions, not an exhaustive assertion of historical priority.

### 1.3 Conventions and the logical status of certificates

We write $\zeta^{(r)}(s,a)$ for differentiation in the order $s$, and $\psi^{(r)}(a)$ for differentiation in the argument. Argument derivatives of generalized Stieltjes constants are explicitly indicated. The convention is

$$\zeta(1+u,a)=\frac{1}{u}+\sum_{m\ge 0}\frac{(-1)^m}{m!}\gamma_m(a)u^m,\qquad \gamma_0(a)=-\psi(a).$$

We use $B_1=-1/2$ and $H_N^{(r)}=\sum_{j=1}^{N}j^{-r}$, with $H_0^{(r)}=0$. Multiple zeta values have decreasing summation indices; $\mathrm{Li}_{s_1,\dots,s_\ell}(z)$ places $z^{n_1}$ on the largest index. Powers and logarithms use the analytic branch specified on each domain.

An ordinary sum, an analytic continuation value, and a constant term in a cutoff expansion are identified only after a proof relates them. Changing the cutoff coordinate can change a finite part. Similarly, a germ holomorphic along common rays need not be jointly holomorphic in independent variables. These distinctions explain the contact polynomials and the multivariate counterexamples below.

Every theorem is proved analytically or by finite exact algebra. Rational determinant certificates establish rank on declared formal coordinates; they do not assert arithmetic independence of zeta values. High-precision calculations independently check the formulas, including complex shifts and degenerate limits, but are not interval certificates or substitutes for proof. The Gaussian $S_6$ and revised $S_8$ identities in the manuscript remain conjectural [1].

## 2 Every minimally compensated vertical-lattice derivative

The vertical-lattice report [3] proves a normally convergent sampling formula after Euler–Maclaurin compensation, and computes a single extra Bernoulli term at the first negative-even derivative. Its Questions 1 and 6 ask for all higher minimally compensated derivatives and for the periodic filters that remove their contacts. We answer both questions. The answer is a single polynomial in the logarithm of the cutoff, specified by an elementary generating function. In particular, higher derivative order does not require additional moments of a periodic filter.

### 2.1 The inherited sampling identity and its exact domain

Let $h>0$, $\tau=2\pi/h$, and $\operatorname{Re}a>0$. All powers use the analytic logarithm on the right half-plane. Write

$$w_n=a+\mathrm{i}hn,\qquad q=e^{-\tau},\qquad z=e^{-\tau a}.$$

The generalized Stieltjes constants have the convention

$$\zeta(1+u,w)=\frac{1}{u}+\sum_{m=0}^{\infty}\frac{(-1)^m}{m!}\gamma_m(w)u^m,\qquad \gamma_0(w)=-\psi(w). \tag{2.1}$$

We use $B_1=-1/2$. For $M\ge 0$, define

$$R_M(s,w)=\zeta(s,w)-\frac{w^{1-s}}{s-1}-\frac{1}{2}w^{-s}-\sum_{j=1}^{M}\frac{B_{2j}}{(2j)!}(s)_{2j-1}w^{1-s-2j}, \tag{2.2}$$

$$\mathcal{L}_\nu(q,z)=\sum_{k=1}^{\infty}\frac{z^k}{k^\nu(1-q^k)}, \tag{2.3}$$

$$\begin{aligned}\mathcal{B}_M(s;a,\tau)=\mathcal{L}_{1-s}(q,z)-\frac{1}{2}\operatorname{Li}_{1-s}(z)-\frac{1}{\tau}\operatorname{Li}_{2-s}(z)\\ -\sum_{j=1}^{M}\frac{B_{2j}\tau^{2j-1}}{(2j)!}\operatorname{Li}_{2-s-2j}(z),\end{aligned} \tag{2.4}$$

$$\mathcal{F}_M(s;a,h)=\frac{\tau^s}{\Gamma(s)}\mathcal{B}_M(s;a,\tau). \tag{2.5}$$

The removable value at $s=1$ is understood in $R_M$. Both $R_M(s,w)$ and $\mathcal{F}_M(s;a,h)$ are entire in $s$.

For completeness, the previously established sampling formula is

$$\sum_{n\in\mathbb{Z}}R_M(s,a+\mathrm{i}hn)=\mathcal{F}_M(s;a,h),\qquad \operatorname{Re}s>-2M. \tag{2.6}$$

It includes all fixed order derivatives on compact subsets of that strict half-plane. One direct proof applies Poisson periodization to

$$\frac{t^{s-1}e^{-at}}{\Gamma(s)}\left(\frac{1}{1-e^{-t}}-\frac{1}{t}-\frac{1}{2}-\sum_{j=1}^{M}\frac{B_{2j}}{(2j)!}t^{2j-1}\right)\quad (t>0),$$

extended by zero for $t\le 0$. The density tends to zero at the origin on the stated sufficient half-plane, and its Fourier coefficients are absolutely summable there. The standard Hurwitz expansion [8, Section 25.11] gives the required normal convergence. We credit (2.6) to the preceding report; it is the starting point, not a new claim.

## 2.2 A universal power-lattice cutoff lemma

The following elementary lemma isolates the whole obstruction.

**Lemma 2.1.** *For $a,h$ as above and $u$ in a sufficiently small complex neighborhood of zero, put*

$$P_K(u)=\sum_{n=-K}^{K}(a+\mathrm{i}hn)^{-1-u},\qquad P_\infty^{\mathrm{an}}(u)=\frac{\tau^{1+u}}{\Gamma(1+u)}\operatorname{Li}_{-u}(e^{-\tau a}).$$

*Then, locally uniformly in $u$ and $a$,*

$$P_K(u)=P_\infty^{\mathrm{an}}(u)+\frac{2}{h}\frac{\sin(\pi u/2)}{u}e^{-u\log(hK)}+O\Big(K^{-1-\operatorname{Re}u}\Big). \tag{2.7}$$

*The sine quotient is filled by its removable value. Derivatives of order $j$ of the error at zero are $O(K^{-1}(1+\log K)^j)$.*

*Proof.* For $\operatorname{Re} u > 0$, ordinary periodization of $t^u e^{-at}/\Gamma(1+u)$ gives $\sum_n (a+\mathrm{i}hn)^{-1-u} = P_\infty^{\mathrm{an}}(u)$. The omitted paired tail has the expansion

$$(a+\mathrm{i}hn)^{-1-u} + (a-\mathrm{i}hn)^{-1-u} = -2\sin(\pi u/2)(hn)^{-1-u} + O(n^{-2-\operatorname{Re} u}).$$

Since $\sum_{n>K} n^{-1-u} = K^{-u}/u + O(K^{-1-\operatorname{Re} u})$, subtracting the tail proves (2.7) in that half-plane. To see the uniform continuation through $u = 0$, use the exact tail expression

$$\begin{aligned}(\mathrm{i}h)^{-1-u}\zeta(1+u, K+1-\mathrm{i}a/h)\\ + (-\mathrm{i}h)^{-1-u}\zeta(1+u, K+1+\mathrm{i}a/h).\end{aligned}$$

Its two residues cancel at $u = 0$. Euler–Maclaurin applied to these two Hurwitz functions proves the stated expansion on a full neighborhood. Differentiating that expansion introduces only powers of $\log K$. The sector and parameter estimates are uniform on compact subsets of $\operatorname{Re} a > 0$. □

At $u = 0$, the lemma gives the familiar distinction

$$\lim_{K\to\infty} P_K(0) = \frac{\tau}{2}\coth\frac{\tau a}{2} = \tau\,\mathrm{Li}_0(e^{-\tau a}) + \frac{\tau}{2}.$$

The extra $\tau/2$ is the zeroth member of the entire cutoff generating function in (2.7).

### 2.3 The all-order contact polynomial

For an even nonnegative integer $N$, define

$$M = N/2, \qquad \beta_N = \frac{B_{N+2}}{(N+1)(N+2)}, \qquad A_N(u) = \prod_{j=1}^{N}\left(1 - \frac{u}{j}\right). \tag{2.8}$$

An empty product equals one. Define the polynomial

$$\boxed{C_{N,d}(X) = \frac{\tau\beta_N}{2}\, d!\,[u^d]\left\{uA_N(u)\frac{\sin(\pi u/2)}{\pi u/2}e^{-Xu}\right\}, \qquad d \ge 1.} \tag{2.9}$$

Its degree is $d-1$, with nonzero leading coefficient $(\tau\beta_N/2)d(-1)^{d-1}$.

**Theorem 2.2** (Complete minimal-contact law)**.** *For every even $N \ge 0$, every $d \ge 1$, and $\operatorname{Re} a > 0$,*

$$\boxed{\sum_{n=-K}^{K} \partial_s^d R_{N/2}(s, a+\mathrm{i}hn)\Big|_{s=-N} = \mathcal{F}_{N/2}^{(d)}(-N; a, h) + C_{N,d}(\log(hK)) + O\left(\frac{(1+\log K)^{d-1}}{K}\right).} \tag{2.10}$$

*Here and below the superscript on $\mathcal{F}$ denotes the order derivative. The error is uniform for $a$ in compact subsets of the right half-plane.*

*For odd $N \ge 1$, put $M = (N+1)/2$. Then all the corresponding derivative sums are absolutely convergent, and*

$$\sum_{n\in\mathbb{Z}} \partial_s^d R_M(s, a+\mathrm{i}hn)\Big|_{s=-N} = \mathcal{F}_M^{(d)}(-N; a, h). \tag{2.11}$$

*Proof.* For even $N$, the difference of two successive compensations is exactly

$$R_{N/2}(-N+u,w) - R_{N/2+1}(-N+u,w) = \beta_N u A_N(u) w^{-1-u}. \tag{2.12}$$

Indeed,

$$\frac{B_{N+2}}{(N+2)!}(-N+u)_{N+1} = \beta_N u A_N(u).$$

The oversubtracted series is normally convergent in $\operatorname{Re} u > -2$. Its omitted tail after $d$ derivatives at zero is $O(K^{-2}(1+\log K)^{d-1})$, since its leading coefficient contains the factor $u$. In the initial half-plane $\operatorname{Re} u > 0$, (2.6) and (2.12) also imply

$$\mathcal{F}_{N/2}(-N+u) - \mathcal{F}_{N/2+1}(-N+u) = \beta_N u A_N(u) P_\infty^{\mathrm{an}}(u).$$

Both sides continue holomorphically. Apply Lemma 2.1 to the truncated version of (2.12). Its explicit cutoff contribution is

$$\frac{2\beta_N}{h} A_N(u) \sin(\pi u/2) e^{-u\log(hK)} = \frac{\tau\beta_N}{2}\, u A_N(u) \frac{\sin(\pi u/2)}{\pi u/2} e^{-u\log(hK)}.$$

Taking the $d$th coefficient proves the theorem and the stated error. For odd $N$, the minimally chosen $M$ already satisfies $-N > -2M$, so (2.11) is an immediate application of normal convergence in (2.6). □

This proof also explains why merely differentiating the first-order Bernoulli correction is insufficient: the factor $A_N(u)\sin(\pi u/2)/(\pi u/2)$ itself contributes higher coefficients.

**Corollary 2.3** (Two explicit cutoff conventions)**.** *The constant term of the cutoff expansion in powers of* $\log K$ *is*

$$\mathrm{FP}_{\log K} \sum_{n\in\mathbb{Z}} \partial_s^d R_{N/2}(-N, a+\mathrm{i}hn) = \mathcal{F}_{N/2}^{(d)}(-N;a,h) + C_{N,d}(\log h). \tag{2.13}$$

*The constant term when the large variable is instead the physical radius* $R = hK$ *is*

$$\mathrm{FP}_{\log R} \sum_{|hn|\le R} \partial_s^d R_{N/2}(-N, a+\mathrm{i}hn) = \mathcal{F}_{N/2}^{(d)}(-N;a,h) + C_{N,d}(0). \tag{2.14}$$

*Only positive powers of the specified logarithm are discarded in either definition. Their difference is exactly* $C_{N,d}(\log h) - C_{N,d}(0)$.

*Proof.* Theorem 2.2 supplies a polynomial plus a term that tends to zero. Taking its constant coefficient in the indicated variable proves both formulas. Replacing $R/h$ by its integer part changes the asymptotic by a quantity tending to zero. □

## 2.4 Harmonic and zeta coefficients

Let $Y_j$ denote the complete exponential Bell polynomials, defined by

$$\exp\left(\sum_{r=1}^{\infty} x_r \frac{u^r}{r!}\right) = \sum_{j=0}^{\infty} Y_j(x_1,\dots,x_j)\frac{u^j}{j!}.$$

The Euler product for sine and the finite product in (2.8) give

$$x_1 = -X - H_N,$$
$$x_r = -(r-1)!\left(H_N^{(r)} + \mathbf{1}_{2|r}\, 2^{1-r}\zeta(r)\right), \qquad r \geq 2, \tag{2.15}$$
$$C_{N,d}(X) = \frac{\tau\beta_N}{2}\, d\, Y_{d-1}(x_1, \ldots, x_{d-1}). \tag{2.16}$$

Thus the contact coefficients involve finite harmonic numbers and even zeta values only; no unspecified Laurent coefficient remains. Writing $L = X + H_N$ and $V = H_N^{(2)} + \zeta(2)/2$, the first four are

$$C_{N,1} = \frac{\tau\beta_N}{2}, \qquad C_{N,2} = -\tau\beta_N L,$$
$$C_{N,3} = \frac{3\tau\beta_N}{2}(L^2 - V), \qquad C_{N,4} = 2\tau\beta_N(-L^3 + 3LV - 2H_N^{(3)}). \tag{2.17}$$

For $N = 0$, in particular,

$$C_{0,1} = \frac{\tau}{24}, \qquad C_{0,2}(X) = -\frac{\tau X}{12}, \qquad C_{0,3}(X) = \frac{\tau}{8}\left(X^2 - \frac{\pi^2}{12}\right). \tag{2.18}$$

The constant $-\tau\pi^2/96$ at the third derivative is missed if the sine factor is dropped.

For explicit evaluation of the analytic term in (2.10), define $E_j^{(N)}$ by

$$\exp\left((\log\tau + \gamma - H_N)u + \sum_{r=2}^{\infty} \frac{(-1)^{r+1}\zeta(r) - H_N^{(r)}}{r} u^r\right) = \sum_{j=0}^{\infty} E_j^{(N)} \frac{u^j}{j!}. \tag{2.19}$$

The reciprocal-Gamma factorization then gives, for every $M$,

$$\mathcal{F}_M^{(d)}(-N; a, h) = (-1)^N N!\tau^{-N} d \sum_{j=0}^{d-1} \binom{d-1}{j} E_j^{(N)} \partial_s^{d-1-j} \mathcal{B}_M(-N; a, \tau). \tag{2.20}$$

This coefficient formula is inherited from [3]; the new result supplies its exact minimal-cutoff interpretation. All order derivatives on the right are normally convergent polylogarithm or Lambert–polylogarithm derivatives.

### 2.5 Periodic filters: the mean is the only obstruction

Let $c_n$ have period $p$ and mean $\bar{c} = p^{-1}\sum_{r=0}^{p-1} c_r$. Define

$$\mathcal{F}_{M,c}(s; a, h) = \sum_{r=0}^{p-1} c_r\, \mathcal{F}_M(s; a + \mathrm{i}hr, ph). \tag{2.21}$$

**Theorem 2.4** (All-order filter criterion)**.** *For even $N$ and $d \geq 1$,*

$$\sum_{n=-K}^{K} c_n \partial_s^d R_{N/2}(-N, a + \mathrm{i}hn) = \mathcal{F}_{N/2,c}^{(d)}(-N; a, h) + \bar{c}\, C_{N,d}(\log(hK)) + o(1). \tag{2.22}$$

*If $\bar{c} = 0$, the positive and negative halves both converge ordinarily, locally uniformly in the order parameter near $-N$; every derivative equals the corresponding derivative of (2.21). No finite part is needed. The condition $\bar{c} = 0$ is necessary and sufficient for all contacts to vanish. There are no additional discrete-moment conditions at higher derivative order.*

*Proof.* Decompose $c_n = \bar{c} + (c_n - \bar{c})$. The constant part is Theorem 2.2. Partial sums of the mean-zero part are bounded. Summation by parts, using

$$\partial_s^d R_{N/2}(-N, a + \mathrm{i}hn) = O\left(\frac{(1 + \log|n|)^{d-1}}{|n|}\right)$$

and its first difference $O((1 + \log|n|)^{d-1}/n^2)$, proves convergence of each half. The same argument before differentiation is locally uniform in $\mathrm{Re}(s + N) > -1$. In $\mathrm{Re}(s + N) > 0$ the sums equal (2.21) by absolute convergence and residue-class splitting; analytic continuation proves equality near $s = -N$. Necessity follows already from $C_{N,1} = \tau\beta_N/2 \neq 0$. □

For the alternating filter this yields the particularly simple all-order identity

$$\sum_{n\in\mathbb{Z}} (-1)^n \partial_s^d R_{N/2}(-N, a + \mathrm{i}hn) = 2\mathcal{F}^{(d)}_{N/2}(-N; a, 2h) - \mathcal{F}^{(d)}_{N/2}(-N; a, h). \tag{2.23}$$

The two half-sums specify the summation convention; absolute convergence is not asserted. For every two shifts $a, b$ in the right half-plane, the relative identity

$$\begin{aligned}\sum_{n\in\mathbb{Z}} c_n \left\{\partial_s^d R_{N/2}(-N, a + \mathrm{i}hn) - \partial_s^d R_{N/2}(-N, b + \mathrm{i}hn)\right\} \\ = \mathcal{F}^{(d)}_{N/2,c}(-N; a, h) - \mathcal{F}^{(d)}_{N/2,c}(-N; b, h)\end{aligned} \tag{2.24}$$

is absolutely convergent for every bounded periodic $c$. The leading reciprocal-logarithmic term is independent of the horizontal shift, so subtraction improves it to $O(n^{-2}(1 + \log|n|)^{d-1})$.

## 2.6 Gamma and all Stieltjes antiderivatives

At $N = 0$, write $D_d(w) = \partial_s^d R_0(0, w)$. Direct differentiation of the fixed subtractions gives

$$D_d(w) = \zeta^{(d)}(0, w) + d!\, w \sum_{j=0}^{d} \frac{(-\operatorname{Log} w)^j}{j!} - \frac{1}{2}(-\operatorname{Log} w)^d. \tag{2.25}$$

The first instance is the usual Stirling remainder,

$$D_1(w) = \log\Gamma(w) - (w - \tfrac{1}{2})\operatorname{Log} w + w - \tfrac{1}{2}\log(2\pi).$$

Its vertical Gamma product belongs to the published work of Arabi Ardehali and Rosengren [14], as the preceding report correctly records. Our contribution here is the explicit all-$d$ minimal-cutoff and filter law.

For example,

$$\begin{aligned}\sum_{n=-K}^{K} \left\{\zeta''(0, w_n) + w_n(\operatorname{Log}^2 w_n - 2\operatorname{Log} w_n + 2) - \tfrac{1}{2}\operatorname{Log}^2 w_n\right\} \\ = \mathcal{F}_0''(0; a, h) - \frac{\tau}{12}\log(hK) + O(K^{-1}\log K).\end{aligned} \tag{2.26}$$

This identity is simultaneously a formula for compensated antiderivatives of the first generalized Stieltjes constant: expansion of $\partial_w\zeta(s, w) = -s\zeta(s + 1, w)$ gives

$$\partial_w \zeta^{(d)}(0, w) = (-1)^d d\, \gamma_{d-1}(w), \qquad d \geq 1. \tag{2.27}$$

Thus

$$\mathfrak{A}_m(w) = \frac{(-1)^{m+1}}{m+1}\left(\zeta^{(m+1)}(0,w) - \zeta^{(m+1)}(0,1)\right)$$

is the uniquely anchored primitive $\mathfrak{A}_m'(w) = \gamma_m(w)$, $\mathfrak{A}_m(1) = 0$. Theorem 2.2, (2.25), and (2.20) evaluate every corresponding compensated lattice sum using finite harmonic/zeta coefficients and Lambert–polylogarithm order derivatives. Formula (2.24) supplies an ordinary absolutely convergent version whenever two anchors are compared.

# 3 Exact modular transport of the order derivatives

Question 7 of [3] asks for an exact $\tau \leftrightarrow 4\pi^2/\tau$ transport, including the nonmodular terms. At horizontal shift $a = 1$, we give one holomorphic identity near every nonpositive even order and another near the Stieltjes order $s = 1$. Their derivatives are specified by finite polynomial recurrences and complete residue formulas.

This section uses classical Mellin-transform technology. The eta and Ramanujan specializations, the zeta functional equation, and the existence of modular relations for generalized Stieltjes constants are established literature, not novelty claims. In particular, the all-index modular relations of Dixit, Sathyanarayana and Sharan [15] preclude any claim that modular Stieltjes identities themselves originate here. The contribution to the present continuation is the explicit order-dependent kernel, its residue compiler, and its compatibility with the fixed-cutoff law of Section 2.

## 3.1 One contour identity before coefficient extraction

Put

$$\Lambda(s,\tau) = \sum_{n=1}^{\infty} \frac{n^{s-1}}{e^{n\tau}-1}, \qquad \tau^* = \frac{4\pi^2}{\tau}, \qquad \alpha = \frac{2\pi}{\tau}, \qquad L = \log\alpha. \tag{3.1}$$

The function $\Lambda$ is entire in $s$. It is exactly $\mathcal{L}_{1-s}(e^{-\tau}, e^{-\tau})$, so subtracting the finite polylogarithm terms in (2.4) recovers the vertically sampled family.

For $M \geq 0$, let

$$\mathcal{Q}_M(s,\tau) = \frac{\zeta(2-s)}{\tau} + \Gamma(s)\zeta(s)\tau^{-s} - \frac{1}{2}\zeta(1-s) + \sum_{j=1}^{M+1} \frac{B_{2j}}{(2j)!}\zeta(2-2j-s)\tau^{2j-1}. \tag{3.2}$$

Near $s = 1$, instead put

$$\mathcal{Q}_*(s,\tau) = \frac{\zeta(2-s)}{\tau} + \Gamma(s)\zeta(s)\tau^{-s} - \frac{1}{2}\zeta(1-s). \tag{3.3}$$

The apparently singular terms are combined before their removable values are taken.

**Theorem 3.1** (Local modular transport)**.** *Fix $\tau > 0$. If $M \geq 0$ and $|s+2M| < 1/4$, then*

$$\boxed{\Lambda(s,\tau) = \mathcal{Q}_M(s,\tau) + \frac{\alpha^s}{2\pi\mathrm{i}}\int_{\operatorname{Re} w = 3/2} \Gamma(w)\zeta(w)\zeta(w+1-s)\frac{\cos(\pi w/2)}{\cos(\pi(s-w)/2)}(\tau^*)^{-w}\,\mathrm{d}w.} \tag{3.4}$$

*If $|s-1| < 1/4$, the same identity holds with $\mathcal{Q}_M$ replaced by $\mathcal{Q}_*$. The contour is oriented upward. Both identities are holomorphic through their central orders, and the integral admits every fixed order derivative under its sign.*

*Proof.* For $c > \max(1, \operatorname{Re} s)$, Mellin inversion and absolute convergence give

$$\Lambda(s,\tau) = \frac{1}{2\pi \mathrm{i}} \int_{\operatorname{Re} z = c} \Gamma(z)\zeta(z)\zeta(z+1-s)\tau^{-z}\, \mathrm{d}z. \tag{3.5}$$

In the stated neighborhoods one may take $c = 3/2$ and move the contour to $\operatorname{Re} z = \operatorname{Re} s - 3/2$. Near $s = -2M$, the poles crossed are $z = 1, s, 0$ and $z = 1 - 2j$, $1 \le j \le M+1$. Their residues are exactly (3.2). The negative even poles of Gamma are canceled by the trivial zeros of $\zeta(z)$. Near $s = 1$, only $z = 1, s, 0$ are crossed, giving (3.3).

The two zeta functional equations [8, Section 25.4] imply the meromorphic identity

$$\begin{aligned}\Gamma(z)\zeta(z)\zeta(z+1-s) = (2\pi)^{2z-s}\Gamma(s-z)\zeta(s-z)\zeta(1-z) \\ \times \frac{\cos(\pi(s-z)/2)}{\cos(\pi z/2)}.\end{aligned} \tag{3.6}$$

Substitute $w = s - z$ on the new contour; the resulting integral is the one in (3.4). Stirling's formula gives exponential decay, while the zeta factors have at most polynomial growth on the fixed vertical strips. The trigonometric quotient and all its order derivatives are bounded there on compact parameter sets: the denominator has no zero on the contour in either stated neighborhood. These facts justify the contour motion and differentiated integrals. Away from the central collisions the residue computation is ordinary. The formulas below explicitly remove those collisions, or equivalently one may use the identity theorem and the holomorphic left-hand side. □

At $s = -2M$, the trigonometric quotient is the constant $(-1)^M$. Thus the zeroth derivative has the classical Ramanujan form

$$\Lambda(-2M,\tau) = \mathcal{Q}_M(-2M,\tau) + (-1)^M \alpha^{-2M}\Lambda(-2M,\tau^*). \tag{3.7}$$

For $M = 0$, it becomes the eta transformation

$$\Lambda(0,\tau) - \Lambda(0,\tau^*) = \frac{\zeta(2)}{\tau} + \frac{1}{2}\log\frac{\tau}{2\pi} - \frac{\tau}{24}. \tag{3.8}$$

For example, at $M = 1$,

$$\begin{aligned}\Lambda(-2,\tau) = \frac{\zeta(4)}{\tau} - \frac{\zeta(3)}{2}\left(1 + \frac{\tau^2}{4\pi^2}\right) + \frac{\zeta(2)\tau}{12} + \frac{\tau^3}{1440} \\ - \frac{\tau^2}{4\pi^2}\Lambda(-2,\tau^*).\end{aligned} \tag{3.9}$$

These are useful normalization checks. The positive $\tau^3/1440$ sign in the last polynomial is forced by $B_4\zeta(0) > 0$.

### 3.2 A finite polynomial compiler for every derivative

Define polynomials $p_j(X)$ by

$$p_0(X) = 1, \qquad p_{j+1}(X) = -(1+X^2)p_j'(X) - Xp_j(X). \tag{3.10}$$

They satisfy

$$\frac{1}{\cos v + X \sin v} = \sum_{j=0}^{\infty} p_j(X)\frac{v^j}{j!}. \tag{3.11}$$

Indeed, differentiating a shifted secant proves the recurrence. The first values are

$$p_1 = -X, \qquad p_2 = 1 + 2X^2, \qquad p_3 = -5X - 6X^3, \qquad p_4 = 5 + 28X^2 + 24X^4.$$

For $M, j, \ell \geq 0$, set

$$\mathcal{T}_{j,\ell}^{(M)}(x) = \frac{(-1)^\ell}{2\pi \mathrm{i}} \int_{\operatorname{Re} w = 3/2} \Gamma(w)\zeta(w)\zeta^{(\ell)}(w + 1 + 2M) p_j\left(\tan \frac{\pi w}{2}\right) x^{-w} \, \mathrm{d}w. \tag{3.12}$$

Every integral is absolutely convergent. For $j = 0$,

$$\mathcal{T}_{0,\ell}^{(M)}(x) = \partial_s^\ell \Lambda(s, x)\Big|_{s=-2M}.$$

**Corollary 3.2** (All negative-even modular derivatives)**.** *For every $r \geq 0$,*

$$\boxed{\partial_s^r \Lambda(-2M, \tau) = \mathcal{Q}_M^{(r)}(-2M, \tau) + (-1)^M \alpha^{-2M} \sum_{\substack{j,\ell,k \geq 0 \\ j+\ell+k=r}} \frac{r!}{j!\ell!k!} \left(\frac{\pi}{2}\right)^j L^k \mathcal{T}_{j,\ell}^{(M)}(\tau^*).} \tag{3.13}$$

*Proof.* Put $s = -2M + u$. The quotient in (3.4) becomes

$$\frac{(-1)^M}{\cos(\pi u/2) + \tan(\pi w/2)\sin(\pi u/2)}.$$

Expand this by (3.11), $\alpha^u = e^{Lu}$, and $\zeta(w + 1 + 2M - u)$. Normal convergence allows coefficient extraction. The multinomial coefficient is the ordinary Leibniz coefficient. □

The terms $j = 0$ are the familiar transformed Lambert derivatives. The finitely many $j \geq 1$ terms are the exact nonmodular correction. Omitting them is already wrong at the first derivative. This is an identity with explicitly specified ordinary contour integrals, not an assertion that those integrals always reduce to a finite list of polylogarithm values.

The corresponding critical-order compiler is equally explicit. Define

$$\mathcal{U}_{j,\ell}(x) = \frac{(-1)^\ell}{2\pi \mathrm{i}} \int_{\operatorname{Re} w = 3/2} \Gamma(w)\zeta(w)\zeta^{(\ell)}(w) X p_j(-X) x^{-w} \, \mathrm{d}w, \qquad X = \cot \frac{\pi w}{2}. \tag{3.14}$$

Since the trigonometric quotient at $s = 1 + u$ is

$$\frac{X}{\cos(\pi u/2) - X \sin(\pi u/2)},$$

we obtain

$$\partial_s^r \Lambda(1, \tau) = \mathcal{Q}_*^{(r)}(1, \tau) + \alpha \sum_{j+\ell+k=r} \frac{r!}{j!\ell!k!} \left(\frac{\pi}{2}\right)^j L^k \mathcal{U}_{j,\ell}(\tau^*). \tag{3.15}$$

Thus every critical-order derivative is transported by finitely many integrals involving $\zeta(w)\zeta^{(\ell)}(w)$.

### 3.3 The colliding residues and their Stieltjes coefficients

The finite residue terms in the preceding formulas require care. At zero, let

$$e_j = \frac{\mathrm{d}^j}{\mathrm{d}u^j}\left(e^{Lu}\sec\frac{\pi u}{2}\right)\Bigg|_{u=0}. \tag{3.16}$$

Then $e_0 = 1$, $e_1 = L$, $e_2 = L^2 + \pi^2/4$, and $e_3 = L^3 + 3\pi^2 L/4$. The zeta functional equation gives

$$\Gamma(s)\zeta(s)\tau^{-s} - \frac{1}{2}\zeta(1-s) = \frac{1}{2}\zeta(1-s)\left(\alpha^s \sec\frac{\pi s}{2} - 1\right).$$

The factor in parentheses vanishes at zero. Multiplying the full Laurent series, including its pole, yields

$$\begin{aligned}\mathcal{Q}_0^{(r)}(0,\tau) = (-1)^r &\left(\frac{\zeta^{(r)}(2)}{\tau} + \frac{\tau}{12}\zeta^{(r)}(0)\right)\\ &+ \frac{1}{2}\left\{-\frac{e_{r+1}}{r+1} + \sum_{j=0}^{r-1}\binom{r}{j}\gamma_j e_{r-j}\right\}.\end{aligned} \tag{3.17}$$

The sum is empty at $r = 0$. In particular, the residue at derivative order $r$ needs one extra coefficient $e_{r+1}$. Replacing the Laurent germ by its regular part would delete this term.

For $M \geq 1$ there is no Laurent collision at $s = -2M$. The same $e_j$, with $L = \log\alpha$, give every residue derivative without separately differentiating a singular Gamma factor:

$$\begin{aligned}\mathcal{Q}_M^{(r)}(-2M,\tau) = (-1)^r &\left\{\frac{\zeta^{(r)}(2M+2)}{\tau} + \sum_{j=1}^{M+1}\frac{B_{2j}\tau^{2j-1}}{(2j)!}\zeta^{(r)}(2M+2-2j)\right\}\\ &+ \frac{1}{2}\left\{(-1)^M\alpha^{-2M}\sum_{\ell=0}^{r}\binom{r}{\ell}(-1)^\ell\zeta^{(\ell)}(2M+1)e_{r-\ell} - (-1)^r\zeta^{(r)}(2M+1)\right\}.\end{aligned} \tag{3.18}$$

Indeed the functional-equation expression immediately preceding (3.17) is a product of two holomorphic functions at these orders. Ordinary Leibniz differentiation proves (3.18).

At $s = 1$, let

$$a_j = \frac{\mathrm{d}^j}{\mathrm{d}u^j}\left(\Gamma(1+u)\tau^{-u}\right)\Big|_{u=0}. \tag{3.19}$$

These are Bell polynomials generated by

$$\Gamma(1+u)\tau^{-u} = \exp\left(-(\gamma+\log\tau)u + \sum_{m=2}^{\infty}\frac{(-1)^m\zeta(m)}{m}u^m\right).$$

The two colliding terms at $s = 1+u$ combine as

$$\frac{1}{\tau}\left\{\frac{A(u)-1}{u} + \sum_{j=0}^{\infty}\frac{\gamma_j u^j}{j!}\left(1+(-1)^j A(u)\right)\right\}, \qquad A(u) = \Gamma(1+u)\tau^{-u}.$$

Consequently,

$$\boxed{\mathcal{Q}_*^{(r)}(1,\tau) = \frac{1}{\tau}\left\{\frac{a_{r+1}}{r+1} + \gamma_r + \sum_{j=0}^{r}\binom{r}{j}(-1)^j\gamma_j a_{r-j}\right\} - \frac{(-1)^r}{2}\zeta^{(r)}(0).} \tag{3.20}$$

This is a finite formula in ordinary Stieltjes constants, zeta values, and logarithms for every $r$. The first two cases are

$$\mathcal{Q}_*(1,\tau) = \frac{\gamma - \log\tau}{\tau} + \frac{1}{4}, \tag{3.21}$$

$$\mathcal{Q}'_*(1,\tau) = \frac{\log^2\tau - \gamma^2 + \zeta(2)}{2\tau} + \frac{1}{2}\zeta'(0). \tag{3.22}$$

Cancellation of $\gamma_1$ in the latter identity is exact.

### 3.4 A self-dual integral evaluation

At $\tau = 2\pi$, $M = 0$, and $r = 1$, the Lambert derivatives on the two sides of (3.13) cancel. Using (3.17) gives the explicit identity

$$\boxed{\frac{1}{2\pi\mathrm{i}}\int_{\operatorname{Re} w=3/2} \Gamma(w)\zeta(w)\zeta(w+1)\tan\frac{\pi w}{2}\,(2\pi)^{-w}\,\mathrm{d}w = -\frac{\zeta'(2)}{\pi^2} - \frac{\pi}{8} + \frac{\log(2\pi)}{6}.} \tag{3.23}$$

The integral is absolutely convergent; no principal value occurs. Higher self-dual derivatives give finite linear identities among the integrals $\mathcal{T}^{(0)}_{j,\ell}(2\pi)$, with complete Stieltjes coefficients supplied by (3.17).

Equations (3.4)–(3.15), together with the finite subtractions in (2.4), answer the modular-transport question at $a = 1$ in the indicated order neighborhoods. General horizontal shifts, and finite polylogarithmic reductions of the correction integrals, remain separate questions.

## 4 A polarized Bernoulli calculus for arbitrary mixed even tails

The all-power continuation in *Bernoulli Tails and Spectral Faces* proves its recurrence for powers of one even-order Hurwitz tail and explicitly leaves products containing unequal orders among three or more factors outside its scope [4, Section 2, after the even-order extension]. The earlier mixed-moment theorem treats two factors [5, Section 4]. We now remove both restrictions: the tail orders may be arbitrary positive even integers, and the number of factors is unrestricted. The result is one finite triangular recurrence, an anchored analytic generating function, and a fixed finite arithmetic space for all regular centered moments.

The unweighted sum of an arbitrary product of zeta tails, and its ordered-index interpretation in multiple zeta values, are classical [9]. Those sums are used below as explicit anchors. The additional assertion concerns *every* centered polynomial weight, including weights requiring polynomial subtraction. The proof keeps the cutoff finite until its boundary terms have cancelled. In particular, this is a continuation of a specified repository question; we do not claim that the classical tail-product anchors are new.

### 4.1 Definitions and ordinary realization of the regular moments

Let $I$ be a nonempty finite set of labels, let $d = |I|$, and assign an even integer $p_i \geq 2$ to each $i \in I$. Equal assigned integers retain their different labels. Write

$$P_S = \sum_{i\in S} p_i, \qquad w_I = P_I - d, \qquad T_p(n) = \zeta(p, n+1) = \zeta(p) - H_n^{(p)}, \qquad F_I(n) = \prod_{i\in I} T_{p_i}(n),$$

and put $F_\varnothing(n) = 1$, $x_n = n + 1/2$. The polygamma normalization is $T_p(n) = \psi^{(p-1)}(n+1)/(p-1)!$ for even $p$.

For $j \geq 0$ define

$$\eta_{p,0} = \frac{1}{p-1}, \qquad \eta_{p,j} = \frac{(p)_{2j-1}}{(2j)!} B_{2j}(1/2) \quad (j \geq 1), \qquad h_{I,j} = \sum_{\substack{j_i \geq 0 \\ \sum_i j_i = j}} \prod_{i \in I} \eta_{p_i, j_i}. \tag{4.1}$$

The centered Hurwitz asymptotics [8, Section 25.11] give

$$F_I(n) \sim \sum_{j \geq 0} h_{I,j} x_n^{-w_I - 2j}. \tag{4.2}$$

Every $\eta_{p,j}$ has sign $(-1)^j$, including the positive $j = 0$ coefficient. Consequently $h_{I,j}$ is nonzero with the same sign. This is an asymptotic expansion, not a convergent Laurent series at a fixed finite $n$.

**Proposition 4.1** (Regular moments as convergent sums)**.** *The Dirichlet series*

$$\mathcal{F}_I(s) = \sum_{n \geq 0} F_I(n) x_n^{-s}, \qquad \Re s > 1 - w_I,$$

*continues meromorphically to the plane. Its poles are precisely $1 - w_I - 2j$, with residue $h_{I,j}$, for $j \geq 0$. For every integer $m \geq 0$ satisfying $m \equiv d \pmod 2$, its regular value $M_{I,m} = \mathcal{F}_I(-m)$ is the ordinary absolutely convergent sum*

$$\boxed{M_{I,m} = \sum_{n=0}^{\infty} \left\{ x_n^m F_I(n) - \sum_{0 \leq j \leq (m - w_I)/2} h_{I,j} x_n^{m - w_I - 2j} \right\}.} \tag{4.3}$$

*A sum with a negative upper limit is empty.*

*Proof.* Subtract finitely many terms of (4.2) from the defining series and add back the corresponding Hurwitz functions $h_{I,j}\zeta(s + w_I + 2j, 1/2)$. Increasing the subtraction order produces normally convergent remainders in successively larger half-planes. The nonzero coefficients prove that all the listed poles are actual. Since $w_I \equiv d \pmod 2$, none of these poles is $-m$ on the stated parity. Every subtracted term in (4.3) contributes $\zeta(-2r, 1/2) = 0$, $r \geq 0$. After the subtraction the summand is $O(x_n^{-2})$, or decays faster when no subtraction is needed. This proves both the value and its ordinary convergence. □

For a full asymptotic expansion in the integer cutoff $N$, let $\mathrm{CT}_N$ mean the coefficient of $N^0 (\log N)^0$. For any integer $k$ put

$$C_{I,k} = \mathrm{CT}_N \left\{ \sum_{n=1}^{N} n^k \big(F_I(n-1) - F_I(n)\big) + N^k F_I(N) \right\}. \tag{4.4}$$

These coefficients exist by asymptotic subtraction: each tail has an expansion in integral inverse powers of $N$, and finite sums of such powers have power–logarithm expansions with a constant coefficient. The convention for this coefficient is fixed by (4.4).

The centered Faulhaber polynomial is

$$P_m(N) = \sum_{n=0}^{N-1} (n + 1/2)^m = \frac{B_{m+1}(N + 1/2) - B_{m+1}(1/2)}{m+1}.$$

Finite summation by parts gives

$$\sum_{n=0}^{N-1} x_n^m F_I(n) = P_m(N)F_I(N) + \sum_{n=1}^{N} P_m(n)\big(F_I(n-1) - F_I(n)\big).$$

All the removed even-power sums in (4.3) are odd polynomials in $N$, with zero constant coefficient. Therefore

$$M_{I,m} = \sum_{k=1}^{m+1} \frac{\binom{m+1}{k}}{m+1} B_{m+1-k}(1/2) C_{I,k}. \tag{4.5}$$

Only indices $k \equiv d+1 \pmod 2$ occur. Define initially as formal series

$$G_I(t) = \sum_{\substack{m\geq 0 \\ m\equiv d \ (2)}} M_{I,m}\frac{t^m}{m!}, \qquad H_I(t) = 2\sinh(t/2)G_I(t), \qquad G_\varnothing = H_\varnothing = 0. \tag{4.6}$$

The Faulhaber generating function makes (4.5) equivalent to

$$H_I(t) = \sum_{\substack{k\geq 1 \\ k\equiv d+1 \ (2)}} C_{I,k}\frac{t^k}{k!}. \tag{4.7}$$

No interchange with a divergent infinite moment sum is used.

## 4.2 The finite polarized identity and the exact boundary parity

For a polynomial in independent variables $a_i$, write $\mathscr{D} = \sum_i h_i \partial_{a_i}$. On a multilinear polynomial this operator is nilpotent. The scalar Bernoulli identity

$$e^z - 1 = \frac{z}{2}(e^z+1) - \sum_{r\geq 1} \frac{B_{2r}}{(2r)!} z^{2r}(e^z - 1)$$

can therefore be applied to $\prod_{i\in I} a_i$ as a finite polynomial identity. It is equivalent to the usual generating function $(z/2)\coth(z/2) = 1 + \sum_{r\geq 1} B_{2r} z^{2r}/(2r)!$.

**Lemma 4.2** (Polarized Bernoulli difference)**.** *If $\Delta F_J(n) = F_J(n-1) - F_J(n)$, then for every $n \geq 1$*

$$\boxed{\begin{aligned}\Delta F_I(n) = \frac{1}{2}\sum_{i\in I} n^{-p_i}\big(F_{I\setminus\{i\}}(n-1) + F_{I\setminus\{i\}}(n)\big) \\ - \sum_{\substack{S\subseteq I \\ |S|\geq 2 \ \text{even}}} B_{|S|} n^{-P_S} \Delta F_{I\setminus S}(n).\end{aligned}} \tag{4.8}$$

*Every product on the right contains fewer tail factors.*

*Proof.* Apply the displayed scalar identity with $a_i = T_{p_i}(n)$ and $h_i = n^{-p_i}$. Since $T_{p_i}(n-1) = T_{p_i}(n) + n^{-p_i}$, the exponential of $\mathscr{D}$ implements the simultaneous shift. The identity

$$\frac{\mathscr{D}^r}{r!}\prod_{i\in I} a_i = \sum_{\substack{S\subseteq I \\ |S|=r}} \left(\prod_{i\in S} h_i\right)\prod_{i\notin S} a_i$$

accounts for every coefficient. The term $S = I$ would be the difference of two constants and is zero. Labels, rather than distinct numerical orders, supply the multiplicities when some $p_i$ coincide. □

Let $\mathcal{A}_p(u)$ be the formal expansion of $T_p(N)$ at $u = N^{-1}$:

$$\mathcal{A}_p(u) = \frac{u^{p-1}}{p-1} - \frac{u^p}{2} + \sum_{r\geq 1} \frac{B_{2r}(p)_{2r-1}}{(2r)!} u^{p+2r-1}. \tag{4.9}$$

Put $b_I(k) = [u^k] \prod_{i\in I} \mathcal{A}_{p_i}(u)$, with $b_\varnothing(k) = \delta_{k,0}$, and define the odd formal series $E_i(u) = \mathcal{A}_{p_i}(u) + u^{p_i}/2$.

**Lemma 4.3** (Boundary cancellation for unequal increments)**.** *For $k \equiv d+1 \pmod 2$,*

$$b_I(k) + \frac{1}{2}\sum_{i\in I} b_{I\setminus\{i\}}(k-p_i) + \sum_{\substack{S\subseteq I \\ |S|\geq 2 \text{ even}}} B_{|S|} b_{I\setminus S}(k - P_S) = 0. \tag{4.10}$$

*More precisely, the entire generating expression on the left is*

$$\boxed{\sum_{\substack{S\subseteq I \\ |S| \text{ even}}} B_{|S|}(1/2) u^{P_S} \prod_{i\notin S} E_i(u).} \tag{4.11}$$

*It has parity $d$.*

*Proof.* Let $B_r^+ = B_r(1)$, so $B_1^+ = 1/2$, $B_r^+ = B_r$ for $r \neq 1$, and odd indices above one vanish. Before taking a coefficient the left side of (4.10) is

$$\sum_{S\subseteq I} B^+_{|S|} u^{P_S} \prod_{i\notin S} \mathcal{A}_{p_i}(u).$$

Substitute $\mathcal{A}_{p_i} = E_i - u^{p_i}/2$. For a fixed subset $R$, the coefficient of $u^{P_R} \prod_{i\notin R} E_i$ is

$$\sum_{S\subseteq R} B_{|S|}(1)(-1/2)^{|R|-|S|} = B_{|R|}(1/2),$$

by the Bernoulli translation identity [8, 24.4.12]. Odd $|R|$ vanish. Each $u^{P_R}$ is even and the remaining product has parity $d - |R| \equiv d$. This proves the identity and cancellation. □

This is the point where equal tail orders are unnecessary. Each tail has its own even increment $u^{p_i}$; Bernoulli translation cancels the entire mixed boundary polynomial at once. If $d$ is even, the full subset in (4.10) has a possible term only at $k = P_I$, an even index outside the required odd parity. Thus its inclusion is harmless, although the corresponding difference in (4.8) is omitted.

### 4.3 Every anchor is a specified convergent multiple zeta value

Our convention is

$$\zeta(s_1, \ldots, s_\ell) = \sum_{n_1 > \cdots > n_\ell \geq 1} n_1^{-s_1} \cdots n_\ell^{-s_\ell}.$$

An ordered set partition of $J$ is a sequence $\pi = (B_1, \ldots, B_\ell)$ of disjoint nonempty blocks whose union is $J$. For $r \geq 1$ and nonempty $J$ define

$$\mathcal{T}_{J,r} = \sum_{\pi \models J} \zeta(P_{B_1}, \ldots, P_{B_\ell}, r), \qquad \mathcal{T}_{\varnothing,r} = \zeta(r) \quad (r \geq 2), \tag{4.12}$$

$$\mathcal{S}_{J,r} = 2\mathcal{T}_{J,r} + \sum_{\varnothing \neq U \subseteq J} \mathcal{T}_{J\setminus U, r+P_U}. \tag{4.13}$$

The only use of $J = \varnothing$ below has $r$ even and at least two. For nonempty $J$ and $h \le 0$ put

$$\mathcal{Z}_{J,h} = \sum_{\pi \models J} \zeta(P_{B_1}, \dots, P_{B_{\ell-1}}, P_{B_\ell} - h), \qquad Z_J = \prod_{i \in J} \zeta(p_i), \quad Z_\varnothing = 1. \tag{4.14}$$

All these MZVs are convergent and have positive integer entries with first entry at least two. In particular no isolated $\zeta(1)$ is introduced.

Index ordering, or direct partition according to the indices equal to $n$, proves the ordinary identities

$$\begin{aligned} \mathcal{T}_{J,r} &= \sum_{n \ge 1} n^{-r} F_J(n), \\ \mathcal{S}_{J,r} &= \sum_{n \ge 1} n^{-r} \big(F_J(n-1) + F_J(n)\big), \\ \mathcal{Z}_{J,h} &= \sum_{n \ge 1} n^{h} \Delta F_J(n) = C_{J,h} \quad (h \le 0). \end{aligned} \tag{4.15}$$

The last equality uses $N^h F_J(N) \to 0$. These are instances of the classical tail-index method [9]; their explicit form makes every constant in the new recurrence independently computable. Their respective weights are $P_J + r$, $P_J + r$, and $P_J - h$.

### 4.4 The all-multiorder recurrence

For $m \ge 0$ of the regular parity for a nonempty $J$, write

$$\mathcal{V}_J(m) = \sum_{\substack{0 \le r \le m \\ r \equiv m \ (2)}} \binom{m}{r} 2^{r-m} M_{J,r} = m! [t^m] \big(\cosh(t/2) G_J(t)\big), \qquad \mathcal{V}_\varnothing(m) = 0.$$

For a label $i \in I$, with $J = I \setminus \{i\}$, define

$$\mathcal{W}_i(k) = \begin{cases} \frac{1}{2} \mathcal{S}_{J, p_i - k}, & k < p_i, \\ \mathcal{V}_J(k - p_i) - \frac{1}{2} \delta_{k, p_i} Z_J, & k \ge p_i. \end{cases} \tag{4.16}$$

**Theorem 4.4** (Triangular mixed-tail recurrence)**.** *For every $k \ge 1$ with $k \equiv d + 1 \pmod 2$,*

$$\boxed{C_{I,k} = \sum_{i \in I} \mathcal{W}_i(k) - \sum_{\substack{S \subsetneq I \\ |S| \ge 2 \text{ even}}} B_{|S|} C_{I \setminus S, k - P_S}.} \tag{4.17}$$

*An index $k - P_S \le 0$ means the explicit anchor $\mathcal{Z}_{I \setminus S, k - P_S}$. Together with (4.5), this is a terminating exact algorithm for every regular moment, because every right-side tail product has strictly smaller cardinality.*

*Proof.* Multiply (4.8) by $n^k$, sum through a finite $N$, and then take constant coefficients. When $m = k - p_i < 0$, the symmetric sum is absolutely convergent and equals $\mathcal{S}_{J,-m}$. When $m \ge 0$, changing the index in one of the two finite sums gives

$$\begin{aligned} \sum_{n=1}^{N} n^m \big(F_J(n-1) + F_J(n)\big) = \sum_{n=0}^{N-1} \big((n+1)^m + n^m\big) F_J(n) \\ - \delta_{m,0} Z_J + N^m F_J(N). \end{aligned}$$

The centered polynomial $(x+1/2)^m + (x-1/2)^m$ has the regular parity for $J$, so the constant coefficient of the first term is $2\mathcal{V}_J(m)$. This remains correct for $J = \varnothing$, where that coefficient is zero by finite power summation. The difference terms yield $C_{I\setminus S,k-P_S}$ minus their boundary coefficients. The complete leftover boundary is precisely (4.10) and vanishes. What remains is (4.17). In particular the exceptional endpoint term $-Z_J/2$ at $k = p_i$ has not been discarded. □

Let $I^q$ denote the anchored ordinary $q$-fold primitive

$$(I^q f)(t) = \frac{1}{(q-1)!}\int_0^t (t-u)^{q-1} f(u)\,du.$$

Here the symbol $I^q$ is an integration operator, distinguished from the label set $I$ by its superscript. Define the finite polynomial

$$R_I(t) = \frac{1}{2}\sum_{i\in I}\sum_{\substack{1\le k<p_i\\ k\equiv d+1\ (2)}} \mathcal{S}_{I\setminus\{i\},p_i-k}\frac{t^k}{k!} - \frac{\mathbf{1}_{d\ \mathrm{odd}}}{2}\sum_{i\in I} Z_{I\setminus\{i\}}\frac{t^{p_i}}{p_i!}$$
$$- \sum_{\substack{S\subsetneq I\\ |S|\ge 2\ \mathrm{even}}} B_{|S|}\sum_{\substack{1\le k\le P_S\\ k\equiv d+1\ (2)}} \mathcal{Z}_{I\setminus S,k-P_S}\frac{t^k}{k!}. \tag{4.18}$$

**Theorem 4.5** (Anchored generators, finite arithmetic span, and polylogarithms)**.** *The series in* (4.6) *converge for* $|t| < 2\pi$*, and are given by*

$$\boxed{\begin{aligned} H_I(t) &= R_I(t) + \sum_{i\in I} I^{p_i}\big(\cosh(t/2)G_{I\setminus\{i\}}(t)\big)\\ &\quad - \sum_{\substack{S\subsetneq I\\ |S|\ge 2\ \mathit{even}}} B_{|S|} I^{P_S} H_{I\setminus S}(t), \qquad G_I(t) = \frac{H_I(t)}{2\sinh(t/2)}. \end{aligned}} \tag{4.19}$$

*For fixed $I$, every $M_{I,m}$ lies in the rational span of* 1 *and convergent positive-index MZVs of weight at most*

$$P_I - 1 - \mathbf{1}_{d\ \mathrm{odd}}. \tag{4.20}$$

*The spanning list is finite and independent of $m$. Moreover $H_I$ has a finite representation by polynomials in $t$, MZV constants, and one-variable multiple polylogarithms at $e^{-t}$ with functional depth at most $d-1$. This is a representation bound, not a claim about minimal depth or arithmetic independence.*

*Proof.* Summing (4.17) with $t^k/k!$ gives (4.19) formally. The ranges $k < p_i$, the endpoint $k = p_i$, and the shifted indices $k - P_S \le 0$ give exactly (4.18). The remaining positive shifted indices give the displayed anchored primitives.

Induction on $d$ proves analyticity. All numerators are holomorphic in $|t| < 2\pi$, have parity $d+1$, and vanish to order at least one at even $d$, or at least two at odd $d$. The denominator has a simple zero at zero and no other zero in this disk. Its singularity is therefore removable, and the formal moment series converges there.

Every non-endpoint anchor in $R_I$ has weight $P_I - k$, and its least possible positive $k$ is one at even $d$ and two at odd $d$. An endpoint product has weight $P_I - p_i \le P_I - 2$ and only occurs at odd $d$; expand it into MZVs by ordered set partitions. All other arithmetic constants already belong

to generators on proper subsets. The remaining Taylor coefficients, integrations, and division by $2\sinh(t/2)$ have rational coefficients. This proves the sharper parity-sensitive bound (4.20), as well as finiteness. The list can be redundant.

For the functional statement, start at real $t > 0$, put $z = e^{-t}$, and use

$$\cosh(t/2)G_J(t) = \left(\frac{1}{2} + \frac{z}{1-z}\right) H_J(t).$$

For $L_{\boldsymbol{a}}(t) = \mathrm{Li}_{a_1,\dots,a_\ell}(e^{-t})$ use the convention with $e^{-tn_1}$ on the largest summation index. Maintain the induction invariant $a_1 \geq 2$ for nonempty words. Ordinary integration raises that first index. An explicit unanchored $q$-fold primitive of $t^m z L_{\boldsymbol{a}}(t)/(1-z)$ is

$$(-1)^q \sum_{j=0}^{m} \binom{q+j-1}{j} \frac{m!}{(m-j)!} t^{m-j} L_{q+j,\boldsymbol{a}}(t). \tag{4.21}$$

It follows by differentiation from $L'_{1,\boldsymbol{a}} = -zL_{\boldsymbol{a}}/(1-z)$ and $L'_{r+1,\boldsymbol{a}} = -L_{r,\boldsymbol{a}}$. Here $q = p_i \geq 2$, so the new first index is at least two, and depth increases by at most one.

Subtract the Taylor polynomial of degree $q-1$ at zero from the *assembled* primitive. Endpoint values with first index at least two are convergent MZVs. The only possible logarithm comes from a single first index one. For admissible $\boldsymbol{a}$,

$$\begin{aligned}\lim_{t\to 0^+} \left(L_{1,\boldsymbol{a}}(t) - \zeta(\boldsymbol{a})L_1(t)\right) = &-\sum_{i=1}^{\ell} \zeta(a_1,\dots,a_i,1,a_{i+1},\dots,a_\ell) \\ &-\sum_{i=1}^{\ell} \zeta(a_1,\dots,a_i+1,\dots,a_\ell).\end{aligned}$$

This follows by subtracting $\zeta(\boldsymbol{a})$ inside the outer series and ordering its missing tail; the difference is absolutely summable. For the empty word the difference is zero. The coefficient of $L_1(t)$ in the full endpoint expression is a multiple of $H_J(0)$, which is zero. Every integration constant is consequently an explicit convergent MZV. The singleton generator is a polynomial and has depth zero; induction gives the bound $d-1$. The combined formula is then continued by its proved holomorphic generator to the stated disk. □

**What has been resolved.** The theorem handles every product of unequal positive even-order tails and every regular centered moment. It also sharpens the unrestricted weight bound by one at odd numbers of factors. It neither changes the summation convention nor assumes a conjecture about MZV independence. It does not prove the separate all-unequal-order *bilinear optimal prefix* conjecture: that problem asks for the nonvanishing of different specified coefficient matrices.

# 5 A mixed cubic, a mixed quartic, and two short convergent identities

In this section a subscript lists the tail orders: $H_{224}$ means $H_{(2,2,4)}$, whereas $H_4$ is the generator of one fourth-order tail. Put

$$E_5 = \frac{15}{2}\zeta(5) - 3\zeta(2)\zeta(3) = \zeta(2,3) + \zeta(4,1) + \zeta(5). \tag{5.1}$$

The second expression follows from the classical Euler reductions for odd-weight double zeta values [9, Equations (4), (9)]. The lower generators supplied by Theorem 4.5 are

$$H_2 = -\frac{t^2}{4}, \qquad H_4 = \frac{\zeta(2)t^2}{2} - \frac{t^4}{48}, \tag{5.2}$$

$$H_{22} = 3\zeta(3)t + 2I^2(\cosh(t/2)G_2),$$

$$H_{24} = E_5 t + \frac{\zeta(3)t^3}{4} + I^2(\cosh(t/2)G_4) + I^4(\cosh(t/2)G_2), \tag{5.3}$$

$$H_{222} = 3I^2(\cosh(t/2)G_{22}) - \frac{17}{8}\zeta(4)t^2 - \frac{\zeta(2)t^4}{48} + \frac{t^6}{2880}.$$

As throughout, $G_J = H_J/(2\sinh(t/2))$. These lower formulas are inherited special cases, included to specify the subsequent mixed formulas without unevaluated initial data.

## 5.1 All moments of $T_2^2 T_4$

**Theorem 5.1** (A six-coordinate mixed cubic)**.** *The complete mixed-cubic generator is*

$$\boxed{\begin{aligned} H_{224}(t) &= 2I^2(\cosh(t/2)G_{24}) + I^4(\cosh(t/2)G_{22}) \\ &\quad - \frac{5}{16}\zeta(6)t^2 - \frac{7}{96}\zeta(4)t^4 - \frac{\zeta(2)t^6}{1440} + \frac{t^8}{161280}. \end{aligned}} \tag{5.4}$$

*Consequently every odd moment $M_{224,2r+1}$ belongs to the rational span of*

$$1,\ \zeta(2),\ \zeta(3),\ \zeta(4),\ E_5,\ \zeta(6). \tag{5.5}$$

*The first six odd moments have full rank on these six formal labels. Thus every later odd moment is a rational linear combination of $M_{224,1}, M_{224,3}, \ldots, M_{224,11}$.*

*Proof.* Apply (4.19) to the three labeled orders $(2,2,4)$. Its nonpolynomial terms are

$$2I^2(\cosh G_{24}) + I^4(\cosh G_{22}) - \frac{1}{6}I^4 H_4 - \frac{1}{3}I^6 H_2.$$

Here and only in this intermediate line, cosh abbreviates $\cosh(t/2)$. The anchor polynomial from (4.18) is

$$-\frac{5}{16}\zeta(6)t^2 - \frac{7}{96}\zeta(4)t^4 - \frac{\zeta(2)t^6}{2160}.$$

For example, its quadratic coefficient simplifies using

$$\mathcal{S}_{\{2,2\},2} = 4\zeta(2,2,2) + 2\zeta(4,2) + 2\zeta(2,4) + \zeta(6),$$

$\zeta(2)\zeta(4) = \zeta(2,4) + \zeta(4,2) + \zeta(6)$, and $\zeta(2,2,2) = 3\zeta(6)/16$. The quartic coefficient uses $\zeta(2)^2 = 5\zeta(4)/2$. Inserting the two singleton generators (5.2) gives exactly (5.4). The lower formulas (5.3) then prove the asserted six-coordinate span for every Taylor coefficient.

In the displayed coordinate order, the first six odd moments have coefficient matrix

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 1 & -5/16 \\ 0 & 1/12 & 1 & -7/16 & -1/6 & 5/64 \\ -19/360 & -29/360 & -7/12 & 35/96 & 47/720 & -35/768 \\ 761/20160 & 2257/20160 & 61/120 & -343/768 & -257/5040 & 155/3072 \\ -967/25200 & -71059/302400 & -2419/3360 & 217/256 & 9521/134400 & -381/4096 \\ 298871/3193344 & 3264013/4561920 & 128981/80640 & -9779/4096 & -40063/253440 & 12775/49152 \end{pmatrix}. \tag{5.6}$$

Its exact rational determinant is

$$-\frac{309971}{13243436236800} \neq 0. \tag{5.7}$$

The entries follow by finite Taylor extraction from (5.4); ordinary rational elimination proves (5.7). Both operations are independently replayed in the supplied verifier. Inverting this finite matrix proves the reconstruction assertion. It asserts rank on specified formal coordinates, not arithmetic independence of the evaluated constants. □

For the first and the first subtracted moments, respectively, this gives

$$\sum_{n\geq 0} x_n T_2(n)^2 T_4(n) = E_5 - \frac{5}{16}\zeta(6), \tag{5.8}$$

$$\sum_{n\geq 0}\left\{x_n^5 T_2(n)^2 T_4(n) - \frac{1}{3}\right\} = -\frac{19}{360} - \frac{29}{360}\zeta(2) - \frac{7}{12}\zeta(3) + \frac{35}{96}\zeta(4) + \frac{47}{720}E_5 - \frac{35}{768}\zeta(6). \tag{5.9}$$

The subtraction in the second line is fixed by

$$T_2(n)^2 T_4(n) = \frac{1}{3x_n^5} - \frac{2}{9x_n^7} + \frac{211}{1080x_n^9} - \frac{337}{1260x_n^{11}} + O(x_n^{-13}).$$

In polygamma notation its product is $\psi'(n+1)^2\psi'''(n+1)/6$.

**Corollary 5.2** (A short cubic identity)**.** *The following series is ordinary and absolutely convergent:*

$$\boxed{\sum_{n=0}^{\infty} T_2(n)T_4(n)(1 - x_n T_2(n)) = \frac{5}{16}\zeta(6).} \tag{5.10}$$

*Its summand is $O(x_n^{-6})$.*

*Proof.* The classical unweighted two-tail evaluation gives $\sum_{n\geq 0} T_2(n)T_4(n) = E_5$; it follows directly by sorting the two indices and is also the linear coefficient in (5.3). Subtract (5.8). Since $T_2(n) = x_n^{-1} - (12x_n^3)^{-1} + O(x_n^{-5})$, the stated decay follows directly.

There is also a finite telescoping proof, independent of the moment recurrence. Define

$$J(N) = -\frac{N^2}{2}T_2(N)^2T_4(N) + \frac{1}{6}T_2(N)^3 - \frac{1}{2}T_2(N)T_4(N) - \frac{1}{6}T_6(N).$$

Direct expansion using $T_p(N-1) = T_p(N) + N^{-p}$ gives the exact identity

$$J(N) - J(N-1) = T_2(N-1)T_4(N-1)\big(1 - (N-1/2)T_2(N-1)\big), \qquad N \geq 1. \tag{5.11}$$

Every term in $J(N)$ tends to zero. At the other endpoint,

$$J(0) = \frac{1}{6}\zeta(2)^3 - \frac{1}{2}\zeta(2)\zeta(4) - \frac{1}{6}\zeta(6) = -\frac{5}{16}\zeta(6).$$

Thus finite summation of (5.11) gives (5.10). In particular, neither a conjectural arithmetic relation nor a regularized limiting argument enters this short evaluation. □

A purely rational identity is also available. Put

$$\begin{aligned} P(x) &= x\big(2526903 + 80590275x^2 + 474388272x^4 \\ &\qquad + 753887520x^6 + 353617920x^8 + 42197760x^{10}\big), \\ Q(x) &= 14065920x^6 + 108495360x^4 + 180958272x^2 + \frac{338793712}{7}. \end{aligned}$$

Then

$$\sum_{n=0}^{\infty}\big\{P(x_n)T_2(n)^2T_4(n) - Q(x_n)\big\} = -6199420. \tag{5.12}$$

The six integer coefficients of $P$ multiply the matrix (5.6) to $(-6199420, 0, 0, 0, 0, 0)$. The displayed centered asymptotic coefficients give exactly $Q$; the residual summand is $O(x_n^{-2})$. Thus this identity follows from finite rational arithmetic and an ordinary convergent sum.

## 5.2 All moments of $T_2^3T_4$

Define the convergent, weight-nine anchor

$$A_9 = \sum_{n=0}^{\infty} T_2(n)^3T_4(n) = \sum_{\pi\models\{1,2,3,4\}} \zeta(P_{B_1}, \ldots, P_{B_{\ell-1}}, P_{B_\ell} - 1), \qquad (p_1, p_2, p_3, p_4) = (2, 2, 2, 4). \tag{5.13}$$

The ordered-partition sum is finite; its grouped fifteen-term expansion is included in the exact verification record. The equality is the classical minimum-index tail identity, with the $n = 0$ term included. Set

$$B_7 = \frac{59}{24}\zeta(3)\zeta(4) + \frac{23}{12}\zeta(2)\zeta(5) - \frac{203}{32}\zeta(7). \tag{5.14}$$

**Theorem 5.3** (An eight-coordinate mixed quartic)**.** *The complete even-moment generator is*

$$\boxed{\begin{aligned} H_{2224}(t) = A_9t + B_7t^3 - \frac{E_5t^5}{180} \\ + 3I^2\big(\cosh(t/2)G_{224}\big) + I^4\big(\cosh(t/2)G_{222}\big) \\ - \frac{1}{2}I^4H_{24} - \frac{1}{2}I^6H_{22}. \end{aligned}} \tag{5.15}$$

*Every even moment is therefore in the rational span of the eight specified coordinates*

$$1,\ \zeta(2),\ \zeta(3),\ \zeta(4),\ \zeta(6),\ E_5,\ B_7,\ A_9. \tag{5.16}$$

*The first eight even moments, of orders* $0, 2, \dots, 14$*, have full formal rank eight. Their coefficient determinant in this order is*

$$-\frac{806331520717}{22490691455232000000} \neq 0. \tag{5.17}$$

*Every later even moment is an explicit rational linear combination of these eight moments.*

*Proof.* There are three singleton deletions of order two and one of order four. Among the even proper subsets there are three pairs of type $(2,2)$ and three of type $(2,4)$; multiplication by $B_2 = 1/6$ gives the two coefficients $1/2$ in (5.15). The anchor polynomial has degrees one, three, and five. Its linear coefficient is $A_9$. Its cubic coefficient is the explicit finite combination

$$\begin{aligned} B_7 = {} & \zeta(2,2,2,1) + \frac{1}{2}\zeta(4,2,1) + \frac{1}{2}\zeta(2,4,1) + \frac{1}{2}\zeta(2,2,3) \\ & + \frac{1}{6}\zeta(6,1) + \frac{1}{6}\zeta(4,3) - \frac{1}{12}\zeta(7). \end{aligned} \tag{5.18}$$

For completeness, let

$$Q_7 = \sum_{n\geq 0} T_2(n)^4 = 51\zeta(3)\zeta(4) + 10\zeta(2)\zeta(5) - \frac{301}{4}\zeta(7).$$

This is Hoffman's classical quartic evaluation, after adding the $n = 0$ term to his convention [9, following Corollary 4.4]. Sorting its four tail indices shows

$$24B_7 = Q_7 - 4\zeta(2,5) - 2\zeta(4,3) - 3\zeta(7).$$

Euler's double-zeta evaluations

$$\zeta(2,5) = 10\zeta(7) - 4\zeta(2)\zeta(5) - 2\zeta(3)\zeta(4), \quad \zeta(4,3) = 17\zeta(7) - 10\zeta(2)\zeta(5)$$

give (5.14). Finally, the quintic anchor coefficient is $-(2\zeta(2,3) + \zeta(5))/240 = -E_5/180$. This proves the generator identity and the eight-coordinate span.

The first eight coefficient rows are obtained by Taylor extraction from the displayed finite system (5.2), (5.3), (5.4), and (5.15). The exact record gives all 64 rational entries; the standalone verifier constructs them and computes (5.17) by rational elimination. This is a finite nonvanishing certificate. Since the generator already proves the span for every order, this one exact determinant proves reconstruction of every later moment from the stated prefix. No independence of the actual period values is required. □

The first three even moments, in compact form, are

$$\begin{aligned} M_{2224,0} &= A_9, \\ M_{2224,2} &= E_5 - \frac{5}{16}\zeta(6) + 2B_7 - \frac{A_9}{12}, \\ M_{2224,4} &= \frac{\zeta(2)}{20} + \frac{3}{2}\zeta(3) - \frac{11}{16}\zeta(4) + \frac{1}{16}\zeta(6) - \frac{23}{60}E_5 - B_7 + \frac{7}{240}A_9. \end{aligned} \tag{5.19}$$

All three are ordinary convergent sums without subtraction, since $T_2^3 T_4 = O(x_n^{-6})$.

**Corollary 5.4** (A short mixed quartic–cubic identity)**.** *One has the ordinary convergent evaluation*

$$\boxed{\begin{aligned}\sum_{n=0}^{\infty} T_2(n)^2 T_4(n)\left\{\left(x_n^2+\frac{1}{12}\right)T_2(n)-x_n\right\}\\ =\frac{59}{12}\zeta(3)\zeta(4)+\frac{23}{6}\zeta(2)\zeta(5)-\frac{203}{16}\zeta(7).\end{aligned}} \tag{5.20}$$

*The summand is* $O(x_n^{-8})$.

*Proof.* The series on the left is $M_{2224,2}+M_{2224,0}/12-M_{224,1}$. Equations (5.8) and (5.19) reduce it to $2B_7$, which is the displayed right side. The centered expansion gives

$$\left(x^2+\frac{1}{12}\right)T_2(n)-x=\frac{1}{45x^3}+O(x^{-5}),\qquad x=x_n.$$

Multiplication by $T_2^2T_4=O(x^{-5})$ proves the asserted decay. □

### 5.3 Independent verification and further questions

The exact verifier uses two different constructions. The first implements Theorem 4.4. The second starts from ordinary finite summation by parts. If

$$\mathcal{T}_{I,q}^{\mathrm{cut}}=\mathrm{CT}_N\sum_{n=1}^{N}n^qF_I(n),\qquad \sum_{n=1}^{N}n^q=\sum_a f_{q,a}N^a,$$

then for $q\ge 0$ its independent recurrence is

$$\mathcal{T}_{I,q}^{\mathrm{cut}}=\sum_a f_{q,a}b_I(a)+\sum_{\varnothing\neq S\subseteq I}\left\{\sum_a f_{q,a}\mathcal{T}_{I\setminus S,a-P_S}^{\mathrm{cut}}-\mathcal{T}_{I\setminus S,q-P_S}^{\mathrm{cut}}\right\}. \tag{5.21}$$

For $q<0$ and nonempty $I$, this is the convergent anchor $\mathcal{T}_{I,-q}$. The empty-product cases are zero for $q\ge 0$, Euler's constant for $q=-1$, and $\zeta(-q)$ for $q\le -2$. The finite identity uses $F_q(n-1)=F_q(n)-n^q$, so its cutoff constants have exactly the required convention.

For all multisets of one through four orders in $\{2,4,6\}$, the two routes agree in 306 regular moments through order 17, coefficient by coefficient on unreduced MZV words. Thus no unproved MZV relation is used to make the comparison succeed. Every Euler-constant label cancels. There are another 825 exact boundary-parity checks, 29 checks of the simplified mixed generators, the six- and eight-coordinate determinant certificates, and the rational row in (5.12). The finite cubic telescoper is checked independently as an exact identity in four Laurent-polynomial variables. The verifier needs only the Python standard library; its complete rational outputs accompany the article.

The separate numerical diagnostic evaluates the ordinary sums at 90 decimal digits, with 40 or 64 initial terms and fixed centered asymptotic tail completions. For the four displayed cubic moments the 40-term residuals are between $3.4\times10^{-72}$ and $1.6\times10^{-62}$; the 64-term residuals are below $10^{-76}$. The quartic second-moment relation has residual below $3.4\times10^{-72}$, and the larger rational identity has residual below $1.3\times10^{-65}$. These are numerical diagnostics, not interval certificates. The proofs are the finite identities, the analytic continuation argument, and exact rational elimination.

Several precise continuations remain:

1. Determine the smallest useful formal arithmetic space for a general multiset of even orders, and find a bound for the moment prefix needed to recover it. The examples above have dimensions six and eight, far below the unrestricted weight-based list.

2. Find further low-degree combinations of mixed moments whose largest-weight anchors cancel, as in (5.10) and (5.20). A systematic classification could produce ordinary convergent identities with only single zeta values and their products.

3. Extend the boundary polynomial to common noninteger shifts. The shifted lattice no longer annihilates every nonnegative even power, so the finite endpoint correction must be retained.

4. Treat odd tail orders or spectral-order derivatives. Their centered parity and logarithmic asymptotic terms differ from the hypotheses proved here; simple polarization of the present theorem does not supply those cases.

5. Identify minimal functional multiple-polylogarithm depth under known functional relations. The bound $d-1$ is constructive, and makes no assertion that a smaller expression is impossible.

# 6 Independent nonperiodic shifts: a completed product generator

The ordinary integral

$$\int_0^1 \gamma_m^{(r)}(a+x)\gamma_n^{(t)}(b+x)\,dx, \qquad \Re a, \Re b > 0,$$

has two independent argument shifts. It is a different object from a periodically translated product. Question R8 of *Tail Kernels and Boundary Products* asks for an order-generating formula, especially its antisymmetric first-argument-derivative component [6]. This section supplies such a formula in an explicit function class: finitely many Gauss hypergeometric terms, holomorphic Gamma divided differences, and two geometrically convergent single Hurwitz-zeta series. It also gives a smaller function class for all products of ordinary polygamma functions.

The mechanism extends the classical product-integral method of Shpot and Paris [11]. Their coincident shift is the case $a = b = 1$ before the spectral poles are removed. The logarithmic series occurring in digamma-square integrals is also classical; see Connon [12]. The additions established here are the independent-shift completion as a *joint* holomorphic germ, its explicit antisymmetric derivative, and the finite closure of all ordinary argument derivatives in one logarithmic Hurwitz family. These are claims relative to the inspected project sources; historical priority for every specialization is not asserted.

## 6.1 Unfolding the two-dimensional lattice

For $\Re A, \Re B > 0$ and $\Re(s+t) > 1$, define the elementary incomplete-beta kernel

$$\begin{aligned}\mathcal{B}_{s,t}(A,B) &= \int_0^\infty \frac{dy}{(y+A)^s(y+B)^t} \\ &= \frac{B^{1-s-t}}{s+t-1}\,{}_2F_1\left(s, s+t-1; s+t; 1-\frac{A}{B}\right).\end{aligned} \tag{6.1}$$

The first expression fixes the branches of the second. Thus the formula is regular when $A = B$, with value $A^{1-s-t}/(s+t-1)$.

**Theorem 6.1** (Exact diagonal unfolding)**.** *For $\Re s, \Re t > 1$ and $\Re a, \Re b > 0$,*

$$\int_0^1 \zeta(s, a+x)\zeta(t, b+x)\,dx = \sum_{k=0}^{\infty} \mathcal{B}_{s,t}(a+k, b) + \sum_{k=1}^{\infty} \mathcal{B}_{s,t}(a, b+k). \tag{6.2}$$

*Both sums converge absolutely and locally normally in this domain.*

*Proof.* Expand both Hurwitz functions by their absolutely convergent Dirichlet series. On the lattice half-plane $m \geq n \geq 0$, put $m = n + k$. For each fixed $k \geq 0$, the intervals $n + x$, $0 \leq x \leq 1$, tile the positive real axis; consequently

$$\sum_{n=0}^{\infty} \int_0^1 (x + n + k + a)^{-s}(x + n + b)^{-t}\,dx = \mathcal{B}_{s,t}(a + k, b).$$

The complementary half-plane $n > m \geq 0$ gives the second sum, starting at $k = 1$. This starting index is essential: the lattice diagonal is counted once. Absolute convergence follows by replacing the complex shifts with lower bounds for their real parts and applying the same argument to positive integrands. The bounds are uniform on compact subsets, proving normal convergence. □

Choose integers $K \geq 0$ and $L \geq 1$, and write

$$c = K + a - b, \qquad d = L + b - a. \tag{6.3}$$

They may always be chosen so large that

$$\Re c > |b|, \qquad \Re d > |a|. \tag{6.4}$$

The choices can be kept fixed on any sufficiently small neighborhood of given shifts. Put

$$\Phi_{s,t}(a, b) = \sum_{k=0}^{K-1} \mathcal{B}_{s,t}(a + k, b) + \sum_{k=1}^{L-1} \mathcal{B}_{s,t}(a, b + k), \tag{6.5}$$

where an empty sum is zero. A useful accelerated form of (6.2) is

$$\begin{aligned} I(s, t; a, b) = \Phi_{s,t}(a, b) &+ B(1 - t, s + t - 1)\zeta(s + t - 1, c) \\ &+ B(1 - s, s + t - 1)\zeta(s + t - 1, d) \\ &- \sum_{j=0}^{\infty} \frac{(-1)^j (s)_j b^{j+1-t}}{j!(j + 1 - t)} \zeta(s + j, c) \\ &- \sum_{j=0}^{\infty} \frac{(-1)^j (t)_j a^{j+1-s}}{j!(j + 1 - s)} \zeta(t + j, d), \end{aligned} \tag{6.6}$$

where $I$ denotes the integral on the left of (6.2). Here $B$ without calligraphic type is Euler's beta function. Identity (6.6) is understood meromorphically; its separate terms need not be regular at integral orders.

Indeed, for a fixed $q$ with $\Re q > |b|$, the change of variable $y + b = z$ in (6.1), followed by subtraction of the interval $0 < z < b$, gives

$$\mathcal{B}_{s,t}(q + b, b) = q^{1-s-t} B(1 - t, s + t - 1) - \sum_{j=0}^{\infty} \frac{(-1)^j (s)_j b^{j+1-t}}{j!(j + 1 - t)} q^{-s-j}. \tag{6.7}$$

This first holds where the beta integral and the subtracted interval both converge, and then holds by meromorphic continuation. Summing over $q = c, c+1, \dots$ proves (6.6); the other tail follows by exchange of the two factors. Alternatively one may first subtract the $j = 0$ term and continue the resulting normally convergent remainder. In either derivation, the condition (6.4) gives a geometric majorant for the $j$-series on compact sets avoiding the displayed poles.

### 6.2 Removal of both spectral poles without an iterated limit

Use the standard convention

$$\mathcal{Z}(u,z) = \zeta(1+u,z) - \frac{1}{u} = \sum_{m=0}^{\infty} \frac{(-1)^m \gamma_m(z)}{m!} u^m. \tag{6.8}$$

For $\Re z > 0$ this is entire in $u$. The required ordinary product generator is

$$\mathcal{J}(u,v;a,b) = \int_0^1 \mathcal{Z}(u,a+x)\mathcal{Z}(v,b+x)\,dx. \tag{6.9}$$

We now evaluate it; its definition alone is not the claimed result. The Hurwitz recurrence and argument-derivative formulas used below have the conventions of DLMF, §25.11 [8].

Define the universal Gamma functions

$$\mathcal{Q}(u,v) = \frac{\Gamma(1+u+v)}{\Gamma(1+u)\Gamma(1+v)}, \qquad \mathcal{E}(w) = \frac{\pi w}{\sin \pi w}.$$

The following quotient has a joint removable singularity:

$$\mathcal{T}(u,v) = \frac{u+v-\mathcal{Q}(u,v)\{u\mathcal{E}(v)+v\mathcal{E}(u)\}}{uv(u+v)}. \tag{6.10}$$

For an explicitly holomorphic interpretation, write

$$\mathcal{T}(u,v) = -\frac{\mathcal{Q}(u,v)-1}{uv} - \mathcal{Q}(u,v)\sum_{h=1}^{\infty} e_{2h}\frac{u^{2h-1}+v^{2h-1}}{u+v}, \tag{6.11}$$

$$e_{2h} = 2(1-2^{1-2h})\zeta(2h). \tag{6.12}$$

Each quotient in the series is a polynomial; also $\mathcal{Q}-1$ is divisible by $uv$ in the ring of convergent power series. In particular $\mathcal{T}(0,0) = -2\zeta(2)$.

The only infinite series in the completed answer are

$$\mathcal{V}(u,v;c,z) = -\sum_{j=1}^{\infty} \frac{(-1)^j (1+u)_j z^{j-v}}{j!(j-v)} \zeta(1+u+j,c), \qquad \Re c > |z|. \tag{6.13}$$

They are holomorphic near $(u,v) = (0,0)$, locally normally in all variables. No term at $j = 0$ is included.

**Theorem 6.2** (Completed independent-shift generator)**.** *Under* (6.3)–(6.4)*, the ordinary integral* (6.9) *is the following joint holomorphic germ:*

$$\begin{aligned}\mathcal{J}(u,v;a,b) = {} & \Phi_{1+u,1+v}(a,b) + \mathcal{T}(u,v) \\ & + \frac{b^{-v}\mathcal{Z}(u,c) - \mathcal{Q}(u,v)\mathcal{E}(v)\mathcal{Z}(u+v,c)}{v} \\ & + \frac{a^{-u}\mathcal{Z}(v,d) - \mathcal{Q}(u,v)\mathcal{E}(u)\mathcal{Z}(u+v,d)}{u} \\ & + \mathcal{V}(u,v;c,b) + \mathcal{V}(v,u;d,a).\end{aligned} \tag{6.14}$$

*Every displayed divided difference has its unique holomorphic value on its zero-denominator face. All mixed spectral and argument derivatives may be taken term by term in the two $\mathcal{V}$ series. The result is independent of the permissible integers $K, L$.*

*Proof.* The unit-interval integral of one Hurwitz function is

$$\int_0^1 \zeta(s, a+x)\,dx = \frac{a^{1-s}}{s-1}.$$

Consequently, with $s = 1+u$, $t = 1+v$,

$$\mathcal{J}(u, v; a, b) = I(1+u, 1+v; a, b) - \frac{a^{-u} + b^{-v} - 1}{uv}. \tag{6.15}$$

In (6.6), isolate $j = 0$ in each series and use

$$B(-v, 1+u+v) = -\frac{\mathcal{Q}(u, v)\mathcal{E}(v)}{v}, \qquad B(-u, 1+u+v) = -\frac{\mathcal{Q}(u, v)\mathcal{E}(u)}{u}.$$

Replacing each Hurwitz function with its pole plus (6.8) leaves precisely the two divided differences in (6.14). The remaining rational pole terms combine to (6.10). Formula (6.11) proves that this last apparent singularity is jointly removable, including on $u + v = 0$. The numerator of the first divided difference vanishes identically on $v = 0$; that of the second vanishes on $u = 0$.

For normal convergence, on a compact subset of $\Re c > |z|$ choose $\rho < 1$ with $|z|/\Re c \le \rho$. For large $j$, the defining Hurwitz series gives a bound of the form $C(\Re c)^{-j}$ times a fixed polynomial in $j$; the Pochhammer ratio $(1+u)_j/j!$ also has polynomial growth uniformly for $u$ in a compact neighborhood of zero. Thus the summands in (6.13) are bounded by $C' j^M \rho^j$. Cauchy's estimates on a slightly larger compact set give the same conclusion after any fixed mixed differentiation. The finite prefix is holomorphic because $\Re(2+u+v) > 1$ near the origin. The meromorphic identity therefore extends to a holomorphic equality. Its left side is independent of $K, L$, proving the final assertion. □

**Corollary 6.3** (Every independent-shift Stieltjes product)**.** *For all $m, n, r, t \ge 0$ and $\Re a, \Re b > 0$,*

$$\int_0^1 \gamma_m^{(r)}(a+x)\gamma_n^{(t)}(b+x)\,dx = (-1)^{m+n}\,\partial_a^r \partial_b^t \partial_u^m \partial_v^n \mathcal{J}(u, v; a, b)\Big|_{u=v=0}, \tag{6.16}$$

*where the right side is evaluated by the explicit convergent formula (6.14). All constants of integration and all spectral regularizations are fixed by this formula.*

This is a convergent reduction, not a claim of finite reduction to ordinary zeta values at arbitrary shifts. To obtain a prescribed coefficient, expand the finite Gamma germ through the required degree, use generalized Stieltjes constants at $c, d$, and extract that degree in each geometrically convergent series. The resulting coefficients involve finite harmonic sums from $(1+u)_j$, powers of $\log a$ and $\log b$, and Hurwitz order derivatives. Each requested coefficient therefore has an explicit, rapidly convergent single-sum expression. More explicitly, the entire generalized-harmonic contribution is encoded by the finite-product identity

$$\frac{(1+u)_j}{j!} = \exp\left\{\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} H_j^{(k)} u^k\right\}, \qquad H_j^{(k)} = \sum_{h=1}^{j} h^{-k}, \quad |u| < 1.$$

### 6.3 The finite terms are polylogarithmic at integral orders

There is no need to introduce parameter derivatives of a hypergeometric function as additional undefined constants. For positive $A > B$, put $q = A - B$ and $z = q/A$. Equation (6.1) becomes

$$\mathcal{B}_{1+u,1+v}(A,B) = q^{-1-u-v} \int_0^z \frac{\exp\{(u+v)\log y - v\log(1-y)\}}{1-y}\, dy. \tag{6.17}$$

Every fixed Taylor coefficient is a finite linear combination of harmonic polylogarithms on the alphabet $\{0,1\}$ and logarithms; this follows by the shuffle product for powers of $\log y$ and $\log(1-y)$ followed by one integration with kernel $1/(1-y)$. The coefficient of total degree $N$ has weight at most $N+1$ before multiplication by powers of $\log q$. Other positive $A, B$ follow by exchange of the two factors, and the diagonal is elementary. For example, writing $\ell = \log(A/B)$,

$$\begin{aligned}
\mathcal{B}_{1,1}(A,B) &= \frac{\ell}{A-B}, \\
\partial_u \mathcal{B}_{1+u,1+v}(A,B)\big|_{0,0} &= -\frac{\ell\log A + \mathrm{Li}_2(1-B/A)}{A-B}, \\
\partial_v \mathcal{B}_{1+u,1+v}(A,B)\big|_{0,0} &= -\frac{\ell\log A + \mathrm{Li}_2(1-B/A) - \ell^2/2}{A-B}.
\end{aligned} \tag{6.18}$$

Thus the finite prefix contributes explicitly reducible polylogarithms, whereas the independent-shift information in general remains in the two single zeta series.

### 6.4 An explicit first Stieltjes coefficient

The first spectral coefficient already combines generalized Stieltjes constants, harmonic numbers, dilogarithms, and Hurwitz order derivatives. Define

$$D_1(A,B) = -\frac{\log(A/B)\log A + \mathrm{Li}_2(1-B/A)}{A-B}, \qquad D_1(A,A) = -\frac{1+\log A}{A},$$

and let $\Phi^{[1]}(a,b)$ be the finite prefix (6.5) with every $\mathcal{B}$ replaced by $D_1$. These formulas are real and continuous for all $A, B > 0$.

**Corollary 6.4** (A harmonic–Stieltjes product identity)**.** *For $a, b > 0$, choose $K, L$ satisfying (6.4) and use (6.3). Then*

$$\begin{aligned}
\int_0^1 \gamma_1(a+x)\psi(b+x)\, dx = \Phi^{[1]}(a,b) &+ \zeta(3) - \gamma_2(c) - \frac{1}{2}\gamma_2(d) \\
&+ \log b\, \gamma_1(c) + \zeta(2)\psi(c) + \left\{\zeta(2) - \frac{1}{2}\log^2 a\right\}\psi(d) \\
&- \sum_{j=1}^{\infty} \frac{(-b)^j}{j}\{H_j\zeta(j+1,c) + \zeta'(j+1,c)\} \\
&- \sum_{j=1}^{\infty} \frac{(-a)^j}{j}\left(\frac{1}{j} - \log a\right)\zeta(j+1,d).
\end{aligned} \tag{6.19}$$

*Here $H_j = \sum_{k=1}^{j} 1/k$ and the prime on $\zeta$ denotes its first spectral derivative. Both infinite sums converge geometrically.*

*Proof.* The left side is $\partial_u \mathcal{J}(0,0;a,b)$ under the convention (6.8). The universal term contributes $\partial_u \mathcal{T}(0,0) = \zeta(3)$, and $\partial_u (1+u)_j|_{u=0} = j! H_j$ accounts for the harmonic number. On the face $v = 0$, the first divided difference in (6.14) equals

$$-\{\log b + \psi(1+u) + \gamma\}\mathcal{Z}(u,c) - \partial_u \mathcal{Z}(u,c).$$

Its derivative at zero is $\log b\, \gamma_1(c) + \zeta(2)\psi(c) - \gamma_2(c)$. The derivative of the second divided difference is $\{\zeta(2) - \frac{1}{2}\log^2 a\}\psi(d) - \frac{1}{2}\gamma_2(d)$. The finite prefix follows from (6.18); differentiating the two normally convergent $\mathcal{V}$ series gives the displayed sums. □

### 6.5 A compact logarithmic Hurwitz family

For an integer $p \geq 1$, $c > 0$, $z \geq 0$, set

$$\mathcal{L}_p(c,z) = \sum_{n=0}^{\infty} \frac{\log(1 + z/(n+c))}{(n+c)^p}. \tag{6.20}$$

This is an ordinary absolutely convergent series; it extends holomorphically to a neighborhood of the positive real parameter domain. When $|z| < \Re c$, it has the geometric zeta expansion

$$\mathcal{L}_p(c,z) = -\sum_{j=1}^{\infty} \frac{(-z)^j}{j} \zeta(p+j,c). \tag{6.21}$$

It is convenient notation for an explicit convergent series, not a claim that this logarithmic zeta family is historically new.

Put $R_p = \partial_z \mathcal{L}_p$. Direct partial fractions give the terminating rules

$$R_1(c,z) = \frac{\psi(c+z) - \psi(c)}{z}, \tag{6.22}$$

$$R_p(c,z) = \frac{\zeta(p,c) - R_{p-1}(c,z)}{z} \quad (p \geq 2), \tag{6.23}$$

$$\partial_c \mathcal{L}_p(c,z) = -p\mathcal{L}_{p+1}(c,z) + R_p(c,z) - \zeta(p+1,c). \tag{6.24}$$

All apparent singularities at $z = 0$ are removable. In particular,

$$\mathcal{L}_1(c,z) = \int_0^z \frac{\psi(c+y) - \psi(c)}{y}\, dy, \tag{6.25}$$

with its additive constant fixed by $\mathcal{L}_1(c,0) = 0$. This identifies the precise Gamma primitive needed by the independent shifts. Its derivative is the divided difference of the logarithmic derivative of $\Gamma$.

For completeness, one two-parameter member generates the full family:

$$\sum_{p=1}^{\infty} w^{p-1} \mathcal{L}_p(c,z) = \mathcal{L}_1(c-w, z+w) - \mathcal{L}_1(c-w,w), \qquad |w| < c, \tag{6.26}$$

locally where all logarithms have their anchored branches. Summing the geometric series $\sum_{p\geq 1} w^{p-1}(n+c)^{-p}$ proves this identity. The finite shift rule

$$\mathcal{L}_p(c+1,z) = \mathcal{L}_p(c,z) - c^{-p}\log(1 + z/c) \tag{6.27}$$

makes evaluation at large $z/c$ straightforward: shift $c$ until (6.21) converges geometrically, retaining only finitely many elementary logarithms.

### 6.6 An explicit two-shift digamma product

Define the continuous divided difference

$$D(A,B)=\frac{\log(A/B)}{A-B},\qquad D(A,A)=\frac{1}{A}.$$

For the moment suppose $a,b>0$ and $|a-b|<1$, and abbreviate

$$c=1+a-b,\qquad d=1+b-a.$$

**Theorem 6.5** (Ordinary independent-shift digamma identity)**.**

$$\boxed{\begin{aligned}\int_0^1 \psi(a+x)\psi(b+x)\,dx = {} & D(a,b)+\gamma_1(c)+\gamma_1(d)-2\zeta(2)\\ & +\log b\,\psi(c)+\log a\,\psi(d)\\ & +\mathcal{L}_1(c,b)+\mathcal{L}_1(d,a).\end{aligned}} \tag{6.28}$$

*For arbitrary $a,b>0$, choose $K\geq 0, L\geq 1$ so that the $c,d$ in (6.3) are positive, and replace $D(a,b)$ by*

$$\sum_{k=0}^{K-1} D(a+k,b)+\sum_{k=1}^{L-1} D(a,b+k).$$

*All other terms retain the form in (6.28) with these $c,d$.*

*Proof.* Set $u=v=0$ in (6.14). The first divided difference becomes $\gamma_1(c)-\log b\,\gamma_0(c)$, because both first derivatives of $\mathcal{QE}$ vanish at the origin. The second is the exchanged expression. Also $\mathcal{V}(0,0;c,b)=\mathcal{L}_1(c,b)$, $\gamma_0=-\psi$, and $\mathcal{T}(0,0)=-2\zeta(2)$. This proves the assertion under the geometric conditions. The ordinary logarithmic series extends it to all stated positive parameters. Independence of the cutoffs can alternatively be checked without analytic continuation: increasing $K$ by one adds $D(K+a,b)$ to the prefix, while the shift formulas for $\gamma_1$, $\psi$, and (6.27) subtract exactly that same number. The argument for $L$ is identical. □

The diagonal is particularly transparent:

$$\int_a^{a+1}\psi(x)^2\,dx=\frac{1}{a}-2\gamma\log a+2\gamma_1-2\zeta(2)+2\mathcal{L}_1(1,a),\qquad a>0. \tag{6.29}$$

This specialization belongs to the established digamma-square and logarithmic-series literature [12]; it is included as a normalization check, not relabelled as a new result.

### 6.7 The antisymmetric first-derivative component

Let

$$W_0(A,B)=\frac{2\log(A/B)-(A-B)(A^{-1}+B^{-1})}{(A-B)^2},\qquad W_0(A,A)=0. \tag{6.30}$$

For $a,b>0$, $|a-b|<1$, keep $c=1+a-b, d=1+b-a$.

**Theorem 6.6** (Explicit antisymmetric derivative)**.** *The ordinary integral*

$$W(a,b)=\int_0^1\{\psi(a+x)\psi'(b+x)-\psi'(a+x)\psi(b+x)\}\,dx$$

*has the exact evaluation*

$$\boxed{\begin{aligned}W(a,b)&=W_0(a,b)\\&\quad+2\{\zeta'(2,d)-\zeta'(2,c)+\log a\,\zeta(2,d)-\log b\,\zeta(2,c)\\&\qquad+\mathcal{L}_2(c,b)-\mathcal{L}_2(d,a)\}\\&\quad+\frac{2\psi(c)-\psi(a+1)}{b}+\frac{\psi(b+1)-2\psi(d)}{a}.\end{aligned}}\tag{6.31}$$

*For arbitrary positive shifts, use* (6.3)*, replace* $W_0(a,b)$ *by its two finite-prefix sums, and replace* $\psi(a+1),\psi(b+1)$ *by* $\psi(K+a),\psi(L+b)$ *respectively.*

*Proof.* Apply $\partial_b-\partial_a$ to (6.28). The finite divided difference gives $W_0$. For $c=K+a-b$ one has $(\partial_b-\partial_a)c=-2$, and the corresponding derivative of $d$ is 2. Formula (6.24) gives

$$(\partial_b-\partial_a)\mathcal{L}_1(c,b)=2\mathcal{L}_2(c,b)-R_1(c,b)+2\zeta(2,c),$$

and the exchanged term has the opposite form. The standard identity

$$\gamma_1'(z)=\zeta(2,z)+\zeta'(2,z)$$

follows by differentiating the Hurwitz shift derivative at its spectral pole. It cancels the explicit $\zeta(2,c),\zeta(2,d)$ terms just produced. Finally use (6.22), $c+b=K+a$, and $d+a=L+b$. Every differentiation is justified by normal convergence of the ordinary logarithmic series on compact positive parameter sets. □

The symmetric companion is the elementary endpoint identity

$$\int_0^1\{\psi'(a+x)\psi(b+x)+\psi(a+x)\psi'(b+x)\}\,dx=\frac{\psi(a)}{b}+\frac{\psi(b)}{a}+\frac{1}{ab}.\tag{6.32}$$

Combining (6.31) and (6.32) therefore evaluates each ordered first-derivative product separately. The antisymmetric formula contains information not supplied by integration by parts alone.

## 6.8 Closure under every ordinary argument derivative

Write $q=r+t$. In differentiating (6.28), the only non-elementary series that must remain after $q$ derivatives are

$$q!\{(-1)^r\mathcal{L}_{q+1}(c,b)+(-1)^t\mathcal{L}_{q+1}(d,a)\}.\tag{6.33}$$

All other terms reduce finitely to polygamma values, rational functions, logarithms, and Hurwitz-zeta order derivatives. For $q>0$, even the two generalized Stieltjes constants disappear:

$$\gamma_1^{(q)}(z)=(-1)^{q+1}q!\{\zeta'(q+1,z)+H_q\zeta(q+1,z)\}.\tag{6.34}$$

To prove (6.33), note that the highest remaining $\mathcal{L}$ term in a derivative with respect to $a$ of $\mathcal{L}_p(c,b)$ is $-p\mathcal{L}_{p+1}$, whereas a derivative with respect to $b$ has highest term $p\mathcal{L}_{p+1}$. The remaining

$R_p$ terms terminate by (6.22)–(6.23). Induction gives the displayed coefficient. The other term is exchanged. Equation (6.34) follows by taking the coefficient of the linear spectral term in $\partial_z^q \zeta(1+u,z) = (-1)^q (1+u)_q \zeta(q+1+u,z)$. Thus the rules are a terminating reduction algorithm, rather than an instruction to leave arbitrary derivatives unevaluated.

For example, a completely explicit trigamma-square identity is

$$\int_a^{a+1} \psi'(x)^2\,dx = \frac{1}{3a^3} + 4\zeta'(3) + \frac{4\zeta(2)}{a} + 4\zeta(3)\log a \\ - \frac{2\{\psi(1+a)+\gamma\}}{a^2} - 4\mathcal{L}_3(1,a), \qquad a>0. \tag{6.35}$$

It is obtained by differentiating the two independent shifts first and only then setting $a=b$. Merely differentiating (6.29) twice would yield the sum of several ordered derivative products, not this individual square.

## 6.9 Diagonal limits and the inherited coincident finite parts

There are two different diagonal operations. Setting $a=b>0$ is an ordinary holomorphic specialization of (6.14). Sending the common shift to zero is a boundary operation and requires removal of the local divergences. For fixed indices $m,n,r,t$, write

$$P(x) = \gamma_m^{(r)}(x)\gamma_n^{(t)}(x).$$

The exact recurrence $\gamma_m(x) = x^{-1}\log^m x + \gamma_m(1+x)$ supplies its finite singular expansion

$$P(x) = \sum_{\alpha\ge 1, h\ge 0} C_{\alpha,h} x^{-\alpha}\log^h x + O(1+|\log x|^M)$$

near zero, with finitely many terms in the displayed singular sum. The coefficients are explicit products of the Laurent singular polynomials and Taylor derivatives at one. For each monomial let $A_{\alpha,h}(x)$ be an antiderivative with zero constant term in its power-log expansion. Then

$$\lim_{a\downarrow 0}\left\{\int_0^1 P(a+x)\,dx + \sum_{\alpha,h} C_{\alpha,h}A_{\alpha,h}(a)\right\} = \mathrm{FP}\int_0^1 P(x)\,dx. \tag{6.36}$$

Indeed, the shifted integral is $\int_a^{1+a} P$, and its difference from $\int_a^1 P$ tends to zero. After integration, the singular monomials contribute $-C_{\alpha,h}A_{\alpha,h}(a)$; the remaining integral converges by the displayed local bound. This proves the boundary compatibility for every fixed order, in the original coordinate $x$, without exchanging an unjustified spectral limit with a finite-part operation.

For example, (6.29) and (6.35) give the exact checks

$$\begin{aligned} \mathrm{FP}\int_0^1 \psi(x)^2\,dx &= 2\gamma_1 - 2\zeta(2), \\ \mathrm{FP}\int_0^1 \psi(x)\psi'(x)\,dx &= \zeta(2), \\ \mathrm{FP}\int_0^1 \psi'(x)^2\,dx &= 2\zeta(3) + 4\zeta'(3). \end{aligned} \tag{6.37}$$

These agree with the inherited coincident product generator. They are compatibility checks, not newly claimed coincident evaluations. Unequal-rate collisions $a,b\to 0$ are not covered by a claim that one can omit the local subtractions; their path-dependent corrections remain a separate question.

### 6.10 Source reconciliation and further targets

There is a concrete status correction to the incoming research list. Question R7 of *Tail Kernels and Boundary Products* asks for all nonpositive-order continuations of the alternating square law. The cutoff-jet section of the independently archived *Bilinear Tails and Determinants* already proves the requisite normal subtractions, all cutoff contacts, their spectral jets, the order-zero $1/4$ discrepancy, and the change under a translated cutoff [7]. R7 should therefore be marked resolved by that report, with its stated coordinate convention. Its Gamma-square specializations should not be counted again as new results here.

The completed generator resolves R8 in the explicit convergent function class described above. It does not claim that arbitrary independent shifts reduce to finitely many ordinary zeta constants. Three sharper questions remain:

1. Determine special rational pairs $(a, b)$ for which the two logarithmic Hurwitz series in (6.31) reduce further to a finite collection of classical polylogarithms and Gamma values. The coincident and complementary-shift cases should be separated from generic rational pairs.

2. Derive functional equations in the shift difference $a - b$ that relate finite descriptions of the prefix by polylogarithms to the two geometric zeta series. The exact $K, L$ invariance gives necessary tests for any proposed relation.

3. Compute the local correction for $a = \lambda\epsilon$, $b = \mu\epsilon$ with fixed positive $\lambda, \mu$ and compare its constant term with the coincident finite part. The generator makes this a defined two-parameter collision problem, but the diagonal compatibility theorem alone does not settle it.

## 7 Exact coefficients for every spectral face

The newest incoming spectral-face report gives an all-order common-ray formula and asks, at the end of its spectral section, for complete collision coefficients and independently variable spectral orders [4]. We answer both questions. A further specialization evaluates every face with just one displaced mass by an ordinary series of *depth-two* polylogarithms. The classical Dirichlet integral, Gamma product, and polylogarithm differentiation identities are ingredients, not novelty claims [13, 8].

### 7.1 The inherited face normalization

Let $k \geq 2$, let $b_1, \dots, b_k$ be real, and let

$$\Sigma_k = \{t_i \geq 0 : \ t_1 + \cdots + t_k = 1\}.$$

The measure is $dt_1 \cdots dt_{k-1}$, with the last coordinate eliminated. For a nonempty subset $T \subseteq \{1, \dots, k\}$ put

$$t_T = \sum_{i \in T} t_i, \qquad b_T = \frac{\sum_{i \in T} t_i b_i}{t_T}.$$

The perspective $t_T F(b_T)$ is filled by zero when $t_T = 0$. Define

$$\mathcal{H}_k(F;t) = \sum_{\varnothing \neq T \subseteq \{1,\dots,k\}} (-1)^{k-|T|} t_T F(b_T), \tag{7.1}$$

$$I_k(F; b_1, \dots, b_k) = \int_{\Sigma_k} \mathcal{H}_k(F;t) \frac{dt}{t_1 \cdots t_k}. \tag{7.2}$$

The notation $I_J$ denotes the same construction with the labels in a subset $J$. This is exactly the predecessor's face normalization; in particular no Euclidean surface-area factor is inserted.

For $F \in C^1([\min b_i, \max b_i])$, the integral is ordinary and absolutely convergent. Indeed the perspective is Lipschitz on its nonnegative coordinate cone, because

$$\partial_{t_i}\{t_T F(b_T)\} = F(b_T) + F'(b_T)(b_i - b_T) \quad (i \in T).$$

Terms cancel in pairs when any coordinate is zero. Consequently $|\mathcal{H}_k(F;t)| \le C \min_i t_i$. On a region where one coordinate is maximal, it is at least $1/k$ and $\min_i t_i \le (\prod_{i \neq i_{\max}} t_i)^{1/(k-1)}$. This also proves that

$$\int_{\Sigma_k} \mathcal{H}_k(F;t) \prod_i t_i^{\alpha_i - 1}\, dt \quad \text{is holomorphic if} \quad \operatorname{Re}\alpha_i > -\frac{1}{k-1} \text{ for every } i, \tag{7.3}$$

with locally normally convergent logarithmic derivatives. These elementary estimates justify every face integral used below. Constants and affine functions are annihilated by $\mathcal{H}_k$.

## 7.2 A finite harmonic formula at every polynomial degree

For $m \ge 2$ and $\ell \ge 0$ define

$$\mathfrak{h}_\ell(m) = [v^\ell] \prod_{j=1}^{m-1} (1 - v/j)^{-1}. \tag{7.4}$$

Thus $\mathfrak{h}_\ell(m)$ is the complete homogeneous polynomial of degree $\ell$ in $1, 1/2, \dots, 1/(m-1)$. In the usual generalized harmonic notation, its first values are

$$1, \qquad H_{m-1}, \qquad \frac{H_{m-1}^2 + H_{m-1}^{(2)}}{2}, \qquad \frac{H_{m-1}^3 + 3H_{m-1}H_{m-1}^{(2)} + 2H_{m-1}^{(3)}}{6}.$$

It is also the weakly ordered harmonic sum $H^\star_{m-1}(\{1\}^\ell)$.

**Theorem 7.1** (Every monomial face integral)**.** *Let $k \ge 2$ and $m \ge 2$. For a vector $\boldsymbol{h} = (h_1, \dots, h_k) \in \mathbb{N}_0^k$ let $d(\boldsymbol{h}) = |\{i : h_i > 0\}|$. Then*

$$\boxed{I_k(B \mapsto B^m; \boldsymbol{b}) = m \sum_{h_1 + \cdots + h_k = m} (-1)^{k-d} (k-d)!\, \mathfrak{h}_{k-d}(m) \prod_{h_i > 0} \frac{b_i^{h_i}}{h_i}.} \tag{7.5}$$

*The right side is a finite symmetric polynomial with rational coefficients. The formula extends to complex $b_i$ as a polynomial identity.*

*Proof.* The perspective is homogeneous of degree one. Radial integration therefore gives

$$I_k(F; \boldsymbol{b}) = \int_{(0,\infty)^k} e^{-\sum_i t_i} \mathcal{H}_k(F;t) \frac{dt_1 \cdots dt_k}{\prod_i t_i};$$

the radial factor is $\int_0^\infty e^{-\rho}\,d\rho = 1$. For a monomial of degree $m \geq 2$, use

$$t_T b_T^m = \frac{(\sum_{i\in T} t_i b_i)^m}{t_T^{m-1}}, \qquad t_T^{1-m} = \frac{1}{(m-2)!}\int_0^\infty v^{m-2} e^{-vt_T}\,dv.$$

Fix a monomial vector $\boldsymbol{h}$ with support $U$ and $d = |U|$. The alternating sum over $T \supseteq U$ is

$$e^{-v\sum_{i\in U} t_i} \prod_{j\notin U} (e^{-vt_j} - 1).$$

Integrating the active and inactive coordinates respectively gives

$$\frac{\Gamma(h_i)}{(1+v)^{h_i}}, \qquad \int_0^\infty e^{-t}\frac{e^{-vt}-1}{t}\,dt = -\log(1+v).$$

The last identity follows by differentiating in $v$ and fixing the value at zero. Every resulting term is absolutely integrable: its remaining majorant is a constant times $v^{m-2}(1+v)^{-m}\log^{k-d}(1+v)$. Thus the use of Fubini is legitimate after the finite inclusion–exclusion has been combined. We obtain

$$I_k(B^m;\boldsymbol{b}) = m(m-1)\sum_{|\boldsymbol{h}|=m} (-1)^{k-d} \prod_{h_i>0} \frac{b_i^{h_i}}{h_i} \int_0^\infty \frac{v^{m-2}\log^{k-d}(1+v)}{(1+v)^m}\,dv.$$

On setting $x = (1+v)^{-1}$, the integral becomes

$$\int_0^1 (1-x)^{m-2}(-\log x)^\ell\,dx = \frac{\ell!}{m-1}\mathfrak{h}_\ell(m), \qquad \ell = k-d.$$

To verify the last equality, expand the elementary beta integral

$$\int_0^1 x^u(1-x)^{m-2}\,dx = \frac{1}{m-1}\prod_{j=1}^{m-1}(1+u/j)^{-1}$$

at $u = 0$. Substitution proves the stated formula. □

At $m = 2$, Theorem 7.1 reduces to the predecessor's factorial variance identity

$$I_k(B^2;\boldsymbol{b}) = (-1)^{k-1}(k-2)!\sum_{i<j}(b_i - b_j)^2. \tag{7.6}$$

The new formula supplies every degree beyond that quadratic term. For example,

$$I_3(B^2;0,1,2) = 6, \qquad I_3(B^3;0,1,2) = 18, \qquad I_3(B^4;0,1,2) = \frac{97}{2}, \qquad I_3(B^5;0,1,2) = \frac{245}{2}. \tag{7.7}$$

### 7.3 Complete convergent collision expansions

**Corollary 7.2** (All collision coefficients)**.** *Suppose $F$ is holomorphic on $|z-B| < R$ and let $M = \max_i|\beta_i|$. For $|\epsilon|M < R$,*

$$I_k(F; B+\epsilon\beta_1,\dots,B+\epsilon\beta_k) = \sum_{m=2}^{\infty}\frac{F^{(m)}(B)}{m!}\epsilon^m I_k(B^m;\boldsymbol{\beta}), \tag{7.8}$$

*normally on compact subsets of this disk. Every coefficient is given by the finite harmonic formula (7.5).*

*Proof.* The constant and affine Taylor terms vanish. On each smaller closed disk, the Taylor series and its first derivative converge uniformly. The boundary estimate preceding (7.3) is a continuous $C^1$ estimate and permits termwise integration. Cauchy estimates on an intermediate disk give a geometric majorant, including the harmless linear factor in $m$ from the derivative. □

The Gamma example is especially explicit. For $a > 0$ and $B \geq 0$ put

$$G_a(B) = \log\Gamma(a + \mathrm{i}\sqrt{B}) + \log\Gamma(a - \mathrm{i}\sqrt{B}) - \log(2\pi), \tag{7.9}$$

$$R_m(a; B) = \sum_{n=0}^{\infty} \big((n + a)^2 + B\big)^{-m}. \tag{7.10}$$

The Gamma logarithms are analytic logarithms continued from positive real arguments. The paired function is analytic as a function of $B$ on the disk of radius $a^2 + B_0$ about any $B_0 \geq 0$. The Gamma product, or differentiation of its normally convergent logarithmic difference, gives

$$G_a^{(m)}(B) = (-1)^m (m-1)! R_m(a; B), \qquad m \geq 1. \tag{7.11}$$

Consequently

$$\boxed{I_k(G_a; B + \epsilon\boldsymbol{\beta}) = \sum_{m=2}^{\infty} \frac{(-1)^m \epsilon^m}{m} R_m(a; B) I_k(B^m; \boldsymbol{\beta}), \quad |\epsilon| M < a^2 + B.} \tag{7.12}$$

At $B = 0$ the resolvent is simply $\zeta(2m, a)$. For instance,

$$I_3(G_a; 0, \epsilon, 2\epsilon) = 3\zeta(4, a)\epsilon^2 - 6\zeta(6, a)\epsilon^3 + \frac{97}{8}\zeta(8, a)\epsilon^4 - \frac{49}{2}\zeta(10, a)\epsilon^5 + \cdots, \qquad 2|\epsilon| < a^2. \tag{7.13}$$

The positive integer resolvents themselves have a finite polygamma description. For $B > 0$,

$$R_1(a; B) = \frac{\psi(a + \mathrm{i}\sqrt{B}) - \psi(a - \mathrm{i}\sqrt{B})}{2\mathrm{i}\sqrt{B}}, \qquad R_m(a; B) = \frac{(-1)^{m-1}}{(m-1)!} \partial_B^{m-1} R_1(a; B). \tag{7.14}$$

Their apparent singularities at $B = 0$ are removable.

### 7.4 All logarithmic face moments have a rational compiler

For independent parameters $\alpha_1, \ldots, \alpha_k$ define

$$A = \sum_i \alpha_i, \qquad Q_k(\boldsymbol{\alpha}) = \frac{\Gamma(1 + A)}{\prod_i \Gamma(1 + \alpha_i)}, \qquad K_{k,m}(\boldsymbol{\alpha}; \boldsymbol{b}) = Q_k(\boldsymbol{\alpha}) \int_{\Sigma_k} \mathcal{H}_k(B^m; t) \prod_i t_i^{\alpha_i - 1}\, dt.$$

For an increment $v$, let $\Delta_v f(y) = (f(y + v) - f(y))/v$, with $\Delta_0 f = f'$.

**Theorem 7.3** (Rational normalized face moments)**.** *Let $m \geq 2$ and $f_m(y) = (1 + y)_{m-1}^{-1}$. For $U = \{i : h_i > 0\}$, write $A_U = \sum_{i \in U} \alpha_i$. Then*

$$\boxed{K_{k,m} = m! \sum_{|\boldsymbol{h}| = m} \left( \prod_{i=1}^{k} \frac{b_i^{h_i}}{h_i!} \right) \left( \prod_{i \in U} (1 + \alpha_i)_{h_i - 1} \right) \left( \prod_{j \notin U} \Delta_{\alpha_j} \right) f_m(A_U).} \tag{7.15}$$

*In particular this Gamma-normalized integral is a rational function of all the spectral parameters, holomorphic in a neighborhood of zero. Its Taylor coefficients give all its logarithmically weighted face moments by finite rational algebra. Removing $Q_k$ introduces only the finite zeta coefficients in the Taylor series of* $\log\Gamma$.

*Proof.* Repeat the orthant proof of Theorem 7.1 with powers $t_i^{\alpha_i-1}$. The radial factor is $\Gamma(1+A)$, and the normalization leaves division by $\prod_i \Gamma(1+\alpha_i)$. For $i \in U$ the coordinate integral contributes $(1+\alpha_i)_{h_i-1}(1+v)^{-h_i-\alpha_i}$. For $j \notin U$ it contributes $((1+v)^{-\alpha_j}-1)/\alpha_j$. After $x=(1+v)^{-1}$, the remaining integral is

$$\int_0^1 (1-x)^{m-2} x^{A_U} \prod_{j\notin U} \frac{x^{\alpha_j}-1}{\alpha_j}\, dx.$$

It equals $(m-2)!\,(\prod_{j\notin U}\Delta_{\alpha_j}) f_m(A_U)$ by the ordinary beta integral. This proves the formula first where the coordinate integrations are absolute. The face estimate and the removable finite differences extend both sides holomorphically to zero. At zero, $f_m^{(\ell)}(0) = (-1)^\ell \ell! \mathfrak{h}_\ell(m)/(m-1)!$, which recovers (7.5) independently. □

# 8 A depth-two polylogarithm evaluation of every repeated-mass face

The general collision expansion has a stronger closed consequence when all but one mass agree. It holds at arbitrary nonnegative displacement, beyond the disk used by the collision series.

**Lemma 8.1** (One-dimensional reduction of a repeated-mass face)**.** *Let $F \in C^2([B, B+\delta])$, where $\delta \geq 0$. For $k \geq 2$,*

$$\boxed{I_k(F; \underbrace{B, \dots, B}_{k-1}, B+\delta) = (-1)^{k-1} \int_0^1 \frac{d}{dt}\left\{\frac{F(B+\delta t)-F(B)}{t}\right\} [-\log(1-t)]^{k-1}\, dt.} \tag{8.1}$$

*The quotient is filled by its limiting value at zero; the integral is ordinary and absolutely convergent.*

*Proof.* After translating $B$ and scaling $\delta$, it suffices to check a polynomial $F$. Constants and affine terms vanish on both sides. For $F(t)=t^m$, $m \geq 2$, only the composition supported on the single nonzero mass survives in (7.5); its value is $(-1)^{k-1}(k-1)!\mathfrak{h}_{k-1}(m)$. The right side gives

$$(-1)^{k-1}(m-1)\int_0^1 t^{m-2}[-\log(1-t)]^{k-1}\, dt,$$

which has exactly the same value by the beta coefficient calculation. For general $C^2$ functions, approximate $F''$ uniformly by polynomials and integrate twice, preserving $F(B)$ and $F'(B)$. The face estimate gives continuity of the left side in the $C^1$ norm. On the right,

$$\frac{d}{dt}\frac{F(B+\delta t)-F(B)}{t} = \delta^2 \int_0^1 u F''(B+\delta t u)\, du,$$

so uniform $C^2$ convergence and integrability of every fixed power of $-\log(1-t)$ give continuity there as well. This proves the lemma. □

We use the strict-index double polylogarithm

$$\mathrm{Li}_{1,p}(z) = \sum_{m>n\geq 1} \frac{z^m}{mn^p} = \sum_{m=2}^{\infty} \frac{H_{m-1}^{(p)}}{m} z^m, \qquad |z|<1. \tag{8.2}$$

In particular $\frac{d}{dz}\mathrm{Li}_{1,p}(z) = \mathrm{Li}_p(z)/(1-z)$.

**Theorem 8.2** (Every repeated-mass Gamma face has depth two)**.** *Let $a > 0$, $B \geq 0$, $\delta \geq 0$, and $k \geq 2$. With $G_a$ as in (7.9),*

$$\boxed{I_k(G_a; \underbrace{B, \dots, B}_{k-1}, B+\delta) = (-1)^{k-1}(k-1)! \sum_{n=0}^{\infty} \mathrm{Li}_{1,k-1}\left(\frac{\delta}{(n+a)^2 + B + \delta}\right).} \tag{8.3}$$

*The sum is ordinary and absolutely convergent. It converges normally when $(a, B, \delta)$ ranges over a compact subset of $\{a > 0, B \geq 0, \delta \geq 0\}$. Thus the dimension of the face does not force polylogarithm depth greater than two in this parameter sector.*

*Proof.* The Gamma product gives the normally convergent logarithmic difference

$$G_a(B+\delta t) - G_a(B) = -\sum_{n=0}^{\infty} \log\left(1 + \frac{\delta t}{(n+a)^2 + B}\right). \tag{8.4}$$

For $u \geq 0$, define

$$J_k(u) = \int_0^1 \frac{\log(1+ut) - ut/(1+ut)}{t^2} [-\log(1-t)]^{k-1}\, dt.$$

Its integrand is nonnegative; the apparent singularity at $t = 0$ is removable. Differentiation under the integral is justified on compact $u$-intervals and gives

$$J_k'(u) = u \int_0^1 \frac{[-\log(1-t)]^{k-1}}{(1+ut)^2}\, dt.$$

Set $q = u/(1+u)$ and $y = 1 - t$. Expanding $(1 - qy)^{-2}$, with nonnegative terms, yields

$$J_k'(u) = \frac{(k-1)!}{1+u} \mathrm{Li}_{k-1}\left(\frac{u}{1+u}\right).$$

Since

$$\frac{d}{du} \mathrm{Li}_{1,k-1}\left(\frac{u}{1+u}\right) = \frac{1}{1+u} \mathrm{Li}_{k-1}\left(\frac{u}{1+u}\right)$$

and both functions vanish at zero, we have $J_k(u) = (k-1)! \mathrm{Li}_{1,k-1}(u/(1+u))$. Insert (8.4) into Lemma 8.1; positivity permits Tonelli's theorem, and this scalar identity gives (8.3).

For the convergence assertion, all arguments lie in $[0, q_0]$, where $q_0 = \delta/(a^2 + B + \delta) < 1$. From $H_{m-1}^{(p)}/m \leq 1$ for $p \geq 1$,

$$0 \leq \mathrm{Li}_{1,p}(q) \leq \frac{q^2}{1-q}.$$

The summands are therefore $O((n+a)^{-4})$, uniformly on the stated compact parameter sets. This proves finiteness as well as normal convergence. The case $\delta = 0$ is immediate. □

For $k = 2$ this is the previously known log-square covariance, since $\mathrm{Li}_{1,1}(q) = \frac{1}{2}\log^2(1-q)$. For $k = 3$, the double polylogarithm reduces entirely to classical dilogarithms and trilogarithms:

$$\mathrm{Li}_{1,2}(q) = 2\,\mathrm{Li}_3(1-q) - 2\zeta(3) - \log(1-q)\{\mathrm{Li}_2(1-q) + \zeta(2)\}, \qquad 0 < q < 1. \tag{8.5}$$

The formula follows by differentiating both sides and taking the limit at $q = 0$. It provides an exact classical-polylogarithm lattice evaluation of the first genuinely higher face.

For example, at $a = 1$, $B = 0$, and $\delta = 3/10$,

$$I_3(G_1; 0, 0, 3/10) = 2\sum_{n=1}^{\infty} \mathrm{Li}_{1,2}\left(\frac{3}{10n^2+3}\right) = 0.072789986411173884726955592165416506 4396\ldots. \tag{8.6}$$

The decimal is a diagnostic value, not an interval enclosure.

*Remark* 8.3 (A weighted extension). The same proof applies when $G_a$ is replaced by the first spectral jet $F_1(B)$ of $\sum_{n\geq 0} W(n+a)((n+a)^2+B)^{-s}$ and $\deg W \leq 2$. Its second mass derivative is the ordinary series $\sum W(n+a)((n+a)^2+B)^{-2}$. More explicitly, with $x_n = n + a$,

$$F_1(B+\delta t) - F_1(B) - \delta t F_1'(B) = -\sum_{n=0}^{\infty} W(x_n)\left\{\log\left(1+\frac{\delta t}{x_n^2+B}\right) - \frac{\delta t}{x_n^2+B}\right\}.$$

This twice-anchored sum and its first two $t$ derivatives converge normally on compact displacement intervals, since their summands are $O(x_n^{\deg W-4})$. Equality follows from the common second derivative and the two zero initial values at $t = 0$. Every face annihilates the two displayed affine terms. The right side of (8.3) then acquires the factor $W(n+a)$. Its convergence follows from $W(n+a) = O(n^2)$. At higher polynomial degree, additional subtractions are necessary and are not silently included in this statement.

# 9 Independent orders: the exact polar obstruction and mixed jets

The common-ray family can be holomorphic at zero for every fixed ray even when the joint family is not a holomorphic germ. This is a material distinction in extending the predecessor's theorem.

## 9.1 A complete holomorphy criterion

Fix $a > 0$, $r \geq 2$, masses $B_i \geq 0$, and a polynomial $W(x) = \sum_{d=0}^{D} w_d x^d$. Put $x_n = n + a$, $A = \sum_i \alpha_i$, and initially, for $\mathrm{Re}\, A > (D+1)/2$, set

$$\mathscr{Z}_W(\boldsymbol{\alpha}) = \sum_{n=0}^{\infty} W(x_n) \prod_{i=1}^{r} (x_n^2 + B_i)^{-\alpha_i}. \tag{9.1}$$

All quadratic factors are positive, and their powers use real logarithms. Define the finite polynomials

$$C_h(\boldsymbol{\alpha}) = [v^h] \prod_{i=1}^{r} (1 + B_i v)^{-\alpha_i}, \qquad P_W(\boldsymbol{\alpha}) = \sum_{h\geq 1} w_{2h-1} C_h(\boldsymbol{\alpha}). \tag{9.2}$$

Only $h \leq (D+1)/2$ occurs in $P_W$.

**Theorem 9.1** (The complete joint polar part)**.** *Near $\boldsymbol{\alpha} = 0$ there is a holomorphic function $\mathscr{H}_W$ such that*

$$\boxed{\mathscr{Z}_W(\boldsymbol{\alpha}) = \frac{P_W(\boldsymbol{\alpha})}{2A} + \mathscr{H}_W(\boldsymbol{\alpha}), \qquad \mathscr{H}_W(0) = \sum_{d=0}^{D} w_d \zeta(-d, a).} \tag{9.3}$$

*The quotient is a prescribed polar subtraction; its part divisible by $A$ is of course holomorphic. The joint spectrum is holomorphic at zero if and only if either $W$ is even or all the masses $B_i$ are equal.*

*Proof.* Choose a finite head so that $B_i/x_n^2 < \rho < 1$ on the tail. A finite binomial expansion with an analytic remainder gives

$$\prod_i (x_n^2 + B_i)^{-\alpha_i} = x_n^{-2A} \sum_{h=0}^{K} C_h(\boldsymbol{\alpha}) x_n^{-2h} + \text{remainder}.$$

Take $K$ sufficiently large that the weighted remainder and all required parameter derivatives converge normally near zero; any $K$ with $2K + 2 > D + 1$ leaves a strict convergence margin after a small neighborhood is chosen. The removed terms are finite sums of $w_d C_h(\boldsymbol{\alpha})\zeta(2A + 2h - d, a + N)$, plus the retained head. Only $2h - d = 1$ gives a pole through the origin, and its residue is exactly $w_d C_h/(2A)$. Subtracting these terms proves (9.3). At zero, $C_0 = 1$ and $C_h(0) = 0$ for $h \geq 1$; the finite head and tail then give the displayed value of $\mathscr{H}_W(0)$.

If $W$ is even, the polar polynomial vanishes. If every mass is $B$, then $C_h = (-1)^h (A)_h B^h/h!$, which is divisible by $A$ for $h \geq 1$. These prove sufficiency. Conversely, suppose $w_{2H-1}$ is the highest nonzero odd coefficient and $B_i \neq B_j$. Restrict to $\alpha_i = t$, $\alpha_j = -t$, and all other parameters zero. Then $A = 0$ and

$$\prod_\ell (1 + B_\ell v)^{-\alpha_\ell} = \exp\{t[\log(1 + B_j v) - \log(1 + B_i v)]\}.$$

The coefficient of $t^H$ in $C_H$ is $(B_j - B_i)^H/H!$. Lower $C_h$ have degree at most $h < H$ and cannot cancel it. Thus $P_W$ is not divisible by $A$, proving necessity. □

For a fixed vector $\boldsymbol{\lambda}$ with $L = \sum_i \lambda_i \neq 0$, every straight ray has the finite limit

$$\lim_{t \to 0} \mathscr{Z}_W(t\boldsymbol{\lambda}) = \sum_d w_d \zeta(-d, a) + \frac{1}{L} \sum_i \lambda_i p_W(B_i), \qquad p_W(B) = \sum_{h \geq 1} \frac{(-1)^h w_{2h-1}}{2h} B^h. \tag{9.4}$$

This follows from the linear coefficient $C_h(t\boldsymbol{\lambda}) = t(-1)^h \sum_i \lambda_i B_i^h/h + O(t^2)$. For example, $a = 1$, $W = x$, and $(B_1, B_2) = (0, 1)$ give

$$\lim_{t \to 0} \mathscr{Z}_x(t, t) = -\frac{1}{3}, \qquad \lim_{t \to 0} \mathscr{Z}_x(t, 2t) = -\frac{5}{12}. \tag{9.5}$$

Even agreement of all straight-ray limits does not suffice. For $W = x^3 + x/2$ and $(B_1, B_2) = (0, 1)$, the polar term is $\alpha_2^2/(4(\alpha_1 + \alpha_2))$. Every non-tangential straight ray has the same zero-order limit, but along $\alpha_2 = t$, $\alpha_1 = -t + t^3$ this term is $1/(4t)$. The joint germ is therefore singular. These are obstructions to an unqualified extension, not errors in the correctly scoped common-ray theorem of the predecessor.

**Corollary 9.2** (Periodic mean cancellation)**.** *If a bounded periodic weight $c_n$ is inserted in every sum, with mean $\bar{c}$, the polar term in (9.3) is multiplied by $\bar{c}$. The joint germ is holomorphic exactly when $\bar{c} = 0$, or $W$ is even, or all masses agree.*

*Proof.* For a period $q$, the auxiliary Dirichlet series is $q^{-z} \sum_{v=0}^{q-1} c_v \zeta(z, (a + v)/q)$ and its residue at $z = 1$ is $\bar{c}$. All other parts of the proof are unchanged. □

## 9.2 The holomorphic mixed defect after an explicit subtraction

Write

$$E_W(s; B) = \sum_{n \geq 0} W(x_n)(x_n^2 + B)^{-s}, \qquad F_j(B) = \partial_s^j E_W(s; B)\Big|_{s=0}.$$

The one-mass functions are holomorphic at zero and $F_0(B) = \sum_d w_d \zeta(-d, a) + p_W(B)$. Define

$$\mathscr{D}_W(\boldsymbol{\alpha}) = \frac{1}{A} \sum_i \alpha_i E_W(A; B_i) - \mathscr{Z}_W(\boldsymbol{\alpha}). \tag{9.6}$$

Let $p_h = (-1)^h w_{2h-1}/(2h)$, and set

$$\mathscr{L}_W(\boldsymbol{\alpha}) = \sum_{h \geq 1} p_h \left\{ \frac{\sum_i \alpha_i B_i^h}{A} - \frac{h!}{(A)_h} \sum_{h_1 + \cdots + h_r = h} \prod_i \frac{(\alpha_i)_{h_i} B_i^{h_i}}{h_i!} \right\}. \tag{9.7}$$

This is a finite rational function. Its $h = 1$ term is identically zero. In several variables its apparent origin singularities should *not* in general be filled by a common-ray value. For instance,

$$\mathscr{L}_{x^3}(\boldsymbol{\alpha}) = \frac{\sum_{i<j} \alpha_i \alpha_j (B_i - B_j)^2}{4A(1 + A)}. \tag{9.8}$$

**Theorem 9.3** (Independent-order face decomposition)**.** *Put*

$$\mathcal{G}_W(u; B) = \frac{E_W(u; B) - F_0(B)}{u} = \sum_{j \geq 1} \frac{F_j(B)}{j!} u^{j-1}, \qquad Q_J(\boldsymbol{\alpha}) = \frac{\Gamma(1 + \sum_{i \in J} \alpha_i)}{\prod_{i \in J} \Gamma(1 + \alpha_i)}.$$

*The difference $\widetilde{\mathscr{D}}_W = \mathscr{D}_W - \mathscr{L}_W$ is holomorphic near zero and equals*

$$\boxed{\widetilde{\mathscr{D}}_W(\boldsymbol{\alpha}) = - \sum_{\substack{J \subseteq \{1, \ldots, r\} \\ |J| \geq 2}} \left( \prod_{i \in J} \alpha_i \right) Q_J(\boldsymbol{\alpha}) \int_{\Sigma_J} \mathcal{H}_J(\mathcal{G}_W(A; \cdot); t) \prod_{i \in J} t_i^{\alpha_i - 1} \, dt.} \tag{9.9}$$

*A sufficient neighborhood for all displayed integrals and zero-point germs is* $\max_i |\alpha_i| < 1/(4r)$*. The formula supplies ordinary, absolutely convergent integrals and locally normal derivatives there.*

*Proof.* For initially positive real parts, the usual Dirichlet parameterization of the product in (9.1) has total order $A$. Finite subset inversion decomposes the perspective into its faces. Under the Dirichlet measure with parameters $\alpha_i$, the mass $t_J$ has mean $A_J/A$, where $A_J = \sum_{i \in J} \alpha_i$, and is independent of the normalized coordinates within $J$. Thus the contribution of face $J$ is

$$\frac{A_J}{A} \frac{\Gamma(A_J)}{\prod_{i \in J} \Gamma(\alpha_i)} \int_{\Sigma_J} \mathcal{H}_J(E_W(A; \cdot); t) \prod_{i \in J} t_i^{\alpha_i - 1} \, dt = \frac{\prod_{i \in J} \alpha_i}{A} Q_J \int_{\Sigma_J} \mathcal{H}_J(E_W(A; \cdot); t) \prod_{i \in J} t_i^{\alpha_i - 1} \, dt.$$

The singleton terms give the weighted endpoint average in (9.6). Splitting $E_W(A; B) = F_0(B) + A\mathcal{G}_W(A; B)$ yields (9.9); ordinary Dirichlet polynomial moments give exactly (9.7) for the $F_0$ part. The identity continues meromorphically from its common absolute-convergence domain. In the stated polydisk, $|A| < 1/4$, each one-mass zero germ is holomorphic, and (7.3) permits every face integral and its derivatives. This proves the claimed joint holomorphy. □

For even $W$, the local term is zero and both the joint spectrum and the joint defect are holomorphic. More generally, the *raw defect* $\mathscr{D}_W$ is holomorphic at zero precisely when the odd part of $W$ has degree at most one, or all masses agree. With periodic weights, mean zero is an additional sufficient and necessary alternative. Indeed its polar numerator is the negative of the degree-at-least-two part of $P_W$; the highest-degree restriction used in Theorem 9.1 proves necessity. This criterion explains why the spectrum and its defect have different analytic boundaries.

### 9.3 A particularly simple formula for distinct mixed derivatives

For a set $J$ put

$$\Pi_J(\boldsymbol{\alpha};t)=\exp\left\{\sum_{i\in J}\alpha_i\log t_i+\sum_{n\geq 2}\frac{(-1)^n\zeta(n)}{n}\left[\left(\sum_{i\in J}\alpha_i\right)^n-\sum_{i\in J}\alpha_i^n\right]\right\}.$$

This is $Q_J\prod_{i\in J}t_i^{\alpha_i}$. If $\boldsymbol{\nu}$ is a nonnegative multiindex, extraction from (9.9) gives the finite identity

$$\partial^{\boldsymbol{\nu}}\widetilde{\mathscr{D}}_W(0)=-\boldsymbol{\nu}!\sum_{\substack{J\subseteq\operatorname{supp}\boldsymbol{\nu}\\|J|\geq 2}}\sum_{j=1}^{|\boldsymbol{\nu}|-|J|+1}\frac{1}{j!}\int_{\Sigma_J}\mathcal{H}_J(F_j;t)[\boldsymbol{\alpha}^{\boldsymbol{\nu}-\mathbf{1}_J}]\{A^{j-1}\Pi_J(\boldsymbol{\alpha};t)\}\frac{dt}{\prod_{i\in J}t_i}. \tag{9.10}$$

Thus a face contributes only if each of its labels is differentiated at least once. The inner jet has order at most $|\boldsymbol{\nu}|-|J|+1$.

**Corollary 9.4** (Squarefree mixed derivatives)**.** *If $U$ contains $m\geq 2$ distinct labels, then*

$$\boxed{\left(\prod_{i\in U}\partial_{\alpha_i}\right)\widetilde{\mathscr{D}}_W(0)=-\sum_{\substack{J\subseteq U\\|J|\geq 2}}\frac{I_J(F_{m-|J|+1})}{m-|J|+1}.} \tag{9.11}$$

*No logarithmic simplex weights or Gamma Taylor corrections remain. For three labels this is*

$$\partial_{\alpha_1}\partial_{\alpha_2}\partial_{\alpha_3}\widetilde{\mathscr{D}}_W(0)=-\tfrac{1}{2}\{I_{12}(F_2)+I_{13}(F_2)+I_{23}(F_2)\}-I_{123}(F_1).$$

*Proof.* In (9.10), every variable belonging to $J$ is already supplied by the face prefactor. The remaining variables are outside $J$, whereas $\Pi_J$ contains only variables in $J$. Consequently it contributes only its constant coefficient. The required monomial is supplied by $A^{m-|J|}$ and has coefficient $(m-|J|)!$. Division by $(m-|J|+1)!$ gives the stated factor. □

For $\deg W\leq 2$, this has an ordinary logarithmic-sum interpretation. Put $L_i(n)=\log((n+a)^2+B_i)$. With $r\geq 2$ labels,

$$\sum_{n=0}^{\infty}W(n+a)\left\{\frac{1}{r}\sum_{i=1}^{r}L_i(n)^r-\prod_{i=1}^{r}L_i(n)\right\}=(-1)^{r+1}\sum_{\substack{J\subseteq\{1,\dots,r\}\\|J|\geq 2}}\frac{I_J(F_{r-|J|+1})}{r-|J|+1}. \tag{9.12}$$

The leading and first-order changes in $B_i/(n+a)^2$ cancel in the braces. The summand is therefore $O(n^{\deg W-4}\log^{r-2}(2+n))$, and the sum is absolutely convergent. This same estimate locally in the order parameters justifies termwise differentiation of the defect. The local term is zero at these polynomial degrees. Taking the squarefree derivative in the pointwise defect proves (9.12).

### 9.4 Verification and the remaining scope

The supplied exact implementation compares (7.5) against a separately constructed rational Dirichlet-moment formula for $2\leq k\leq 5$ and $2\leq m\leq 8$; all 28 cases agree exactly over the rational numbers. Six additional quadratic variance checks cover $2\leq k\leq 7$.

The numerical program uses 48 decimal digits. It compares direct triangle face integrals of degrees $3,4,6$ against the finite formula; their absolute residuals are respectively approximately

$1.02 \cdot 10^{-26}$, $3.42 \cdot 10^{-25}$, and $1.33 \cdot 10^{-23}$. The repeated-mass one-dimensional formula and the independently expanded positive double-polylogarithm series agree for $k = 2, \dots, 6$ to residuals below $1.7 \cdot 10^{-38}$. Most significantly, the original *two-dimensional Gamma face* in (8.6) was evaluated directly on all six coordinate-ordering sectors. At Gauss–Legendre orders $16, 24, 32$, its residual against the polylogarithm lattice value decreases from $2.33 \cdot 10^{-19}$ to $3.00 \cdot 10^{-28}$ and then $4.41 \cdot 10^{-37}$. The sector map is $t = (1, u, uv)/(1 + u + uv)$ and its weighted measure is $du\, dv/(uv)$. The face numerator is retained as a combined expression throughout.

These computations are finite algebraic checks and numerical diagnostics, not interval certificates or substitutes for the proofs. The results complete the specified all-order collision and independent-order extensions relative to the inspected spectral-face report, and give a global depth-two evaluation in the one-displaced-mass sector. They do not prove a finite classical-polylogarithm reduction for arbitrary distinct masses or every polylogarithm weight. Natural next targets are a finite dictionary for two distinct mass clusters, arithmetic reductions of the resulting double-polylogarithm lattice sums, and a basis-independent description of the joint polar subtraction when the affine spectral factors have different degrees.

# 10 Source reconciliation, verification, and further research

## 10.1 Which earlier questions are settled

The following table records the precise scope of the continuation. The source labels refer to the incoming reports at the pinned snapshot, not to a moving version of the repository.

| **Earlier target** | **Result and remaining scope** |
|---|---|
| Vertical lattice, Questions 1 and 6 [3] | Theorems 2.2 and 2.4 give every minimal contact and the exact periodic mean criterion. All derivative orders are covered. |
| Vertical lattice, Question 7 [3] | Theorem 3.1 gives exact transport for horizontal shift $a = 1$, near every nonpositive even order and near $s = 1$, with every derivative correction. A general arbitrary-shift transport remains open. |
| Unequal products beyond two factors [4, 5] | Theorem 4.5 handles every finite collection of positive even tail orders and every regular centered moment. The separate conjecture about optimal unequal bilinear prefixes remains open. |
| Independent nonperiodic Stieltjes shifts, R8 [6] | Theorem 6.2 supplies an explicit jointly holomorphic convergent generator, including all argument derivatives. A finite reduction in a prescribed classical constant alphabet is a stronger question. |
| Higher spectral collisions and independent orders [4] | The face-moment formula gives all polynomial collision coefficients. The one-displacement Gamma sector is evaluated by depth-two polylogarithms. The joint polar part and its finite local correction are explicit. |
| Alternating nonpositive-order square law, R7 [6] | Already resolved in the separate Bilinear Tails archive [7]; update its status and cite that proof. It is not counted as a new result here. |
| Gaussian $S_6$ and revised $S_8$ [1] | Remain conjectural. The present results do not supply the missing period relations. |

### 10.2 A concrete correction to the research ledger

In the Tail Kernels archive, R7 asks for the alternating square continuation at all nonpositive orders. At the same snapshot, `sections/04_cutoff_jets.tex` in the Bilinear Tails archive already proves the required normally convergent continuation, full contact formula, and translated-cutoff rule [6, 7]. Its theorem labels `cut:normal`, `cut:contacts`, and `cut:translation` locate the exact claims. The order-zero value $1/4$, despite the vanishing pointwise summands in the relevant representation, is explicitly reconciled there. The Gamma specializations have labels `cut:Gamma` and `cut:Gammasquare`.

The proposed replacement ledger entry is:

> **R7: resolved.** Cite the all-order cutoff continuation in *Bilinear Alternating Tails and Harmonic Determinants*, Section 4. Preserve the specified cutoff coordinate and its order-zero contact. Remaining extensions should be stated separately from the already established nonpositive-order law.

This corrects the status shared between two reports. It does not identify a false theorem in the resolving report.

### 10.3 Three safeguards for subsequent integration

**Finite parts must retain their coordinate.** For the new lattice formula, the constant left after deleting positive powers of $\log K$ is $\mathcal{F}_M^{(d)}(-N) + C_{N,d}(\log h)$. Deleting positive powers of the physical-radius logarithm $\log R$, $R = hK$, instead gives $\mathcal{F}_M^{(d)}(-N) + C_{N,d}(0)$. The analytic continuation value is obtained only after the entire contact polynomial has been removed. These are three specified operations; their constant differences should not be suppressed in a shared definition of *regularized sum*.

**Periodic cancellation does not imply absolute convergence.** A nonzero mean-zero periodic multiplier removes the critical contact at every fixed derivative order. The resulting two half-lattice series converge by summation by parts. They are in general conditional, since the terms still have critical power-log magnitudes. An additional horizontal difference improves the decay and gives absolute convergence. Theorem 2.4 and (2.24) keep these claims distinct.

**A common-ray limit is insufficient for independent orders.** The predecessor's common-ray spectral theorem is correctly scoped. An unqualified extension to independent orders would be false. For instance, with $W(x) = x$, $a = 1$, and masses $0, 1$, the limits along $(\alpha_1, \alpha_2) = (t, t)$ and $(t, 2t)$ are $-1/3$ and $-5/12$. Even equality of all nontangential straight-ray limits would not suffice: the example $W(x) = x^3 + x/2$ has polar part

$$\frac{\alpha_2^2}{4(\alpha_1 + \alpha_2)}.$$

It tends to zero on every fixed ray with nonzero total direction, but along $\alpha_2 = t$, $\alpha_1 = -t + t^3$ it equals $1/(4t)$. The joint polar decomposition, rather than a ray calculation, is the appropriate hypothesis for mixed differentiation.

### 10.4 Reproducible verification

The companion files separate exact algebra from numerical diagnostics. All arithmetic determinants and finite polynomial identities are computed over rational or symbolic exact rings. The floating-point checks compare independently organized expressions, including direct quadratures, finite cutoff sums, and Mellin contours.

| Component | Verification supplied |
|---|---|
| Lattice contacts | 60 exact coefficient and Pochhammer checks, two mutation controls, and 27 finite-lattice diagnostics at 55 digits. The next two Euler–Maclaurin coefficients are used to check the remainder when its leading polynomial nearly cancels. |
| Modular transport | Nine exact kernel-polynomial checks, four direct Lambert-versus-contour comparisons at 42 digits, two classical boundary checks, and an independent self-dual contour evaluation. |
| Mixed tails | 306 exact moments for all multisets of one through four orders from $\{2,4,6\}$, through the tested moment range, against finite summation by parts; 825 boundary-parity checks; exact six- and eight-coordinate determinants; a finite telescoping certificate; and 90-digit checks of the short identities. |
| Independent shifts | 16 exact cancellation checks and 42 numerical checks at 65 digits, including complex shifts, both spectral axes, $u+v=0$, prefix invariance, and the explicit $\gamma_1 \times \psi$ coefficient. The largest recorded absolute residual is approximately $1.02 \times 10^{-56}$. |
| Spectral faces | 34 exact rational checks and 11 numerical comparisons at 48-digit working precision. The original two-dimensional Gamma face is checked against the polylogarithmic lattice expression with increasing quadrature orders. |

These calculations are regression diagnostics for the stated proofs. They do not establish global arithmetic independence, replace a normal-convergence argument, or certify decimal enclosures. All domains, branch conventions, and limiting operations used in the identities are established in the text. The verification programs and their recorded outputs are included so that future integration can detect sign, endpoint, normalization, or coefficient regressions.

### 10.5 Further research questions

The following questions are ordered by their connection to the new mechanisms and by the prospect of obtaining further exact identities. Each asks for a structural statement or evaluation.

**Q1. Optimal prefixes for arbitrary unequal tails.** The mixed recurrence proves a finite span, and the two worked products have complete rational prefix certificates. Determine the minimal formal coordinate dimension for a general even-order multiset $I$, after a declared system of MZV reductions, and determine when its first admissible moments already span that space. The existing unequal bilinear optimal-prefix conjecture is a particularly precise first target. A proof should establish an explicit nonzero determinant, rather than infer rank from evaluated decimals.

**Q2. Direct telescoping certificates for elegant tail identities.** The short cubic identity admits a finite telescoping certificate in products of Hurwitz tails. Characterize the centered polynomial combinations for which such a certificate exists. Can the polarized recurrence be compiled into a finite certificate with a prescribed maximal tail order? This could systematically produce short ordinary identities whose cancellations hide a much larger moment system.

**Q3. Minimal functional depth of mixed generators.** The bound $d-1$ is constructive but need not be sharp. Classify the order multisets for which the anchored generator reduces to classical polylogarithms, or to depth two independently of $d$. The classification must distinguish functional identities from numerical MZV reductions at the endpoint $t=0$.

**Q4. More than one displaced spectral mass.** For $k-1$ equal masses and one displaced mass, depth two suffices in every dimension. Determine the exact function class for two displaced masses and, more generally, for a fixed number of distinct mass clusters. Is depth controlled by the number of clusters rather than the number of factors? A useful result would give a finite multiple-polylogarithm formula with explicit branches and ordinary convergence, not merely rename the remaining simplex integral.

**Q5. Complete logarithmic collision coefficients.** The rational normalized face formula produces every logarithmically weighted polynomial moment by differentiation. Find compact closed forms for the first genuinely nonsquarefree mixed order derivatives, and organize their coefficients by harmonic star sums and zeta weight. Determine which Gamma normalization coefficients cancel before any MZV reduction.

**Q6. Independent-order continuation beyond quadratic spectra.** For quadratic factors, the odd powers of $W$ give the exact polar obstruction through a single total-order divisor. Develop the corresponding finite polar-part classification for higher-degree polynomial spectra and shifted linear factors. Identify the natural local subtraction that is compatible with multiplication, parameter differentiation, and periodic mean cancellation.

**Q7. Modular transport with arbitrary horizontal shift.** Extend the exact order-dependent contour identity from $a=1$ to general $a$. Rational shifts should first be organized into finite systems with additive characters and Hurwitz factors. For irrational shifts, determine the additional integral transform explicitly. The fixed-cutoff contact polynomial should remain visible throughout the transport.

**Q8. Arithmetic reductions of the logarithmic Hurwitz family.** For which rational pairs $(c,z)$ does

$$\mathcal{L}_p(c,z)=\sum_{n\geq 0}\frac{\log(1+z/(n+c))}{(n+c)^p}$$

reduce to a finite specified alphabet of polylogarithms, Gamma values, and zeta derivatives? The generating and differential relations in the independent-shift section impose strong compatibility conditions. Generic rational arguments should not be assumed to have such a reduction.

**Q9. Unequal-rate Stieltjes boundary collisions.** Set $a=\lambda\epsilon$, $b=\mu\epsilon$ with fixed $\lambda,\mu>0$, and let $\epsilon\downarrow 0$. Compute the complete singular power-log primitive and the remaining constant for arbitrary Stieltjes indices and argument derivatives. Determine the exact path correction relative to the common-shift finite part. The diagonal unfolding and explicit finite-prefix derivatives provide a starting point.

**Q10. Three independently shifted Stieltjes factors.** Extend the evaluated two-factor germ to three factors with independent shifts and orders. Partitioning the summation lattice into minimum-index regions suggests finite Lauricella-type prefixes. The central problem is to remove all intersecting spectral poles jointly while retaining geometric convergence after every derivative.

**Q11. The unresolved Gaussian period identities.** The manuscript's $S_6$ and revised $S_8$ candidates need exact period relations. The existing high-precision residuals and bounded relation-space calculations do not supply them. Seek a new functional relation or contour deformation that escapes the already tested relation spaces, and exhibit its exact coefficient certificate before numerical evaluation. The new mixed-tail rank results concern different spaces and should not be used as evidence of a Gaussian proof.

**Q12. A formal proof layer for the finite core.** The nilpotent polarized Bernoulli identity, the exact boundary-parity cancellation, the contact-polynomial coefficient identity, and the rational face compiler are natural Lean targets. Separate their finite algebraic statements from Euler–Maclaurin continuation and normal-convergence lemmas. This would make later analytic generalizations depend on a small reusable verified algebraic core, without claiming that the current Python checks constitute formal proofs.

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